

Hierarchical Multilevel Monte Carlo for Order-Optimal Neural Actor-Critic in Average-Reward CMDPs

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Abstract

Constrained Markov Decision Processes (CMDPs) provide a natural framework for reinforcement learning in safety-critical applications, where agents maximize long-term reward while satisfying long-term constraints. Although primal-dual actor-critic methods with linear critics are well understood, extending order-optimal convergence guarantees to neural critics in average-reward CMDPs has remained open. The main challenge is a fundamental bias-cost tradeoff in neural critic estimation: under Neural Tangent Kernel (NTK) analysis, reducing critic bias requires substantially increasing critic optimization cost, preventing order-optimal convergence in the primal-dual framework. We resolve this bottleneck by introducing a hierarchical Multilevel Monte Carlo (MLMC) neural critic that performs debiasing simultaneously across trajectory sampling and critic optimization. The resulting estimator attains the bias of a long critic optimization run with only logarithmic expected sample cost. Building on this estimator, we develop a primal-dual Natural Actor-Critic algorithm that achieves both an optimality gap and a constraint violation of order $\tilde{O}(T^{-1/2})$. This establishes the first order-optimal convergence guarantees for infinite-horizon average-reward CMDPs with general policy parameterization and neural critics, while eliminating the need to know the underlying mixing time. Our results are novel even in the unconstrained setting.

1 Introduction

Reinforcement learning (RL) has emerged as a powerful framework for sequential decision-making, with recent successes largely driven by deep neural networks. Many applications require agents not only to maximize long-term performance but also to satisfy resource or operational constraints, including transportation (Al-Abbasi, Ghosh, and Aggarwal 2019; Haliem et al. 2021), communication networks (Panju et al. 2021), robotics (Chen, Lan, and Aggarwal 2023; Gonzalez et al. 2023), and healthcare (Tamboli et al. 2024). Such problems are naturally modeled as constrained Markov Decision Processes (CMDPs), where the objective is to maximize the long-run average reward while satisfying long-term safety or resource constraints. Compared with discounted formulations, the average-reward criterion directly captures steady-state performance and is therefore particularly suitable for continuing decision-making tasks.

Primal-dual policy-gradient and actor-critic methods have become the dominant framework for solving CMDPs due

to their scalability and compatibility with function approximation. In these methods, the actor updates a parameterized policy using gradient information estimated by a critic, while a dual variable enforces long-term constraints. For linear critics, recent advances have established order-optimal convergence guarantees of $\tilde{O}(T^{-1/2})$ under general policy parameterization (Xu et al. 2025; Mondal and Aggarwal 2024), matching the minimax sample complexity lower bound for average-reward reinforcement learning (Jin and Sidford 2021).

Despite these advances, extending order-optimal guarantees to neural critics remains an important open problem. Neural networks provide substantially greater representational power than linear critics and constitute the standard function approximator in modern reinforcement learning. Recent analyses based on the Neural Tangent Kernel (NTK) have enabled finite-sample guarantees for neural actor-critic methods by showing that, within a neighborhood of the network initialization, neural critics behave approximately as linear models. These techniques have led to order-optimal convergence for unconstrained discounted MDPs (Ganesh et al. 2025). However, for average-reward CMDPs, the best existing result achieves only a $\tilde{O}(T^{-1/4})$ convergence rate (Satheesh et al. 2026), leaving a substantial gap from the optimal $\tilde{O}(T^{-1/2})$ rate attainable with linear critics (Xu et al. 2025).

The difficulty is not merely the use of neural networks, but a fundamental bias-cost tradeoff introduced by neural critic estimation. Under the NTK framework, the projection required to maintain the linear approximation causes the squared critic bias to decay at the same rate as its mean-squared error (MSE). Consequently, reducing the critic bias necessitates running the critic optimization for substantially more iterations. In contrast, the primal-dual actor-critic algorithm operates within a nested-loop architecture, where each outer primal-dual update can only afford a limited amount of critic computation while maintaining the overall sample budget. As a result, the critic MSE remains of order $\tilde{O}(1)$ (this implies $\tilde{O}(1)$ squared critic bias under NTK) whereas order-optimal convergence requires the squared critic bias to decay at a rate $\tilde{O}(T^{-1})$. This creates a fundamental bias-cost bottleneck that prevents existing neural actor-critic algorithms from attaining order-optimal convergence.

In this work, we resolve this bottleneck by introducing a hierarchical Multilevel Monte Carlo (MLMC) neural critic. Existing MLMC techniques randomize trajectory lengths to remove sampling bias under Markovian observations. Such estimators inherit the bias of a long trajectory while incurring only a logarithmic sampling cost. However, they do not address the optimization bias arising from finite neural critic training. Our key insight is to introduce a second MLMC layer that randomizes the optimization horizon itself, thereby debiasing critic optimization in addition to trajectory sampling.

Our main contributions are summarized as follows.

- **Resolving the Neural Critic Bias-Cost Tradeoff:** We identify the fundamental bottleneck preventing order-optimal convergence of primal-dual actor-critic methods with neural critics. To overcome this challenge, we introduce a novel *hierarchical* MLMC neural critic that performs multilayer debiasing across both trajectory sampling and critic optimization. Our estimator inherits the bias of a vanilla critic run for T iterations while requiring only $\mathcal{O}(\log T)$ expected sample cost.
- **First Order-Optimal Guarantees with Neural Critics:** Building upon the proposed hierarchical MLMC neural critic, we establish global convergence and constraint-violation rates of $\tilde{\mathcal{O}}(T^{-1/2})$. To the best of our knowledge, this is the first primal-dual Natural Actor-Critic algorithm with neural critics and general policy parameterization to achieve order-optimal convergence for infinite-horizon CMDPs. Our results are novel even for the unconstrained average-reward setting.
- **Mixing-Time-Free Algorithm:** Unlike existing methods that achieve order-optimal convergence with neural critics, our algorithm requires no knowledge of the mixing time of the underlying CMDP, making it considerably more practical for large-scale reinforcement learning problems.

1.1 Related Works

Average-Reward Constrained MDPs: CMDPs, introduced by (Altman 2021), provide the standard framework for safe reinforcement learning. Recent work has focused on primal-dual policy gradient methods for high-dimensional CMDPs (Ding et al. 2020; Paternain et al. 2019). Initial finite-sample guarantees were established for tabular and linear-function-approximation settings (Xu, Liang, and Lan 2021; Chen, Jain, and Luo 2022; Bai, Bedi, and Aggarwal 2023), while (Agarwal, Bai, and Aggarwal 2022) and (Ghosh, Zhou, and Shroff 2023) achieved optimal convergence with zero constraint violation in model-based and linear model-free settings, respectively. For general policy parameterizations, (Bai, Mondal, and Aggarwal 2024a) established a sub-optimal rate, whereas (Xu et al. 2025) obtained the optimal $\tilde{\mathcal{O}}(T^{-1/2})$ convergence rate.

Actor-Critic Methods with Neural Critics: Early finite-sample analyses of actor-critic methods assumed linear critics (Khodadadian, Chen, and Maguluri 2021; Cayci, He, and Srikant 2024). More recent works leverage Neural Tangent

Kernel (NTK) theory (Jacot, Gabriel, and Hongler 2018) to analyze neural critics in discounted MDPs (Fu, Yang, and Wang 2021; Tian, Olshevsky, and Paschalidis 2023; Cayci, He, and Srikant 2024; Gaur et al. 2024; Ganesh et al. 2025). In particular, (Ganesh et al. 2025) established a $\tilde{\mathcal{O}}(T^{-1/2})$ convergence rate for unconstrained discounted MDPs. For average-reward constrained MDPs, (Satheesh et al. 2026) established a $\tilde{\mathcal{O}}(T^{-1/4})$ convergence rate, which remains the state-of-the-art.

MLMC-Based Estimation: Existing analyses under Markovian sampling often rely on sample thinning (Gaur et al. 2024; Ganesh, Mondal, and Aggarwal 2025a), which requires knowledge of the mixing time. Recent works have instead leveraged Multilevel Monte Carlo (MLMC) methods (Blanchet and Glynn 2015; Beznosikov et al. 2023; Patel et al. 2024; Ganesh, Mondal, and Aggarwal 2025b) to construct bias-controlled estimators via randomized trajectory lengths, thereby eliminating the need for data dropping and mixing-time estimates. In the CMDP setting, (Xu et al. 2025) used MLMC to mitigate the bias-cost tradeoff in primal-dual natural actor-critic algorithms with linear critics.

2 Problem Setup

We consider an infinite-horizon average-reward constrained Markov Decision Process (CMDP) $\mathcal{M} := (\mathcal{S}, \mathcal{A}, \mathcal{P}, r, c, \rho)$, where $\mathcal{S}$ and $\mathcal{A}$ are the state and action spaces, respectively, $\mathcal{P} : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$ denotes the transition kernel,¹ $r : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$ and $c : \mathcal{S} \times \mathcal{A} \rightarrow [-1, 1]$ are reward and cost functions, respectively, and $\rho \in \Delta(\mathcal{S})$ denotes the initial state distribution. A stationary policy $\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$ generates a trajectory $(s_t, a_t)_{t=0}^\infty$, where $s_0 \sim \rho$, $a_t \sim \pi(\cdot|s_t)$, and $s_{t+1} \sim \mathcal{P}(\cdot|s_t, a_t)$.

Throughout this work, we make the following assumption on the underlying CMDP.

Assumption 1 (Ergodicity). *For every policy π , the Markov chain (s_t) induced by π is irreducible and aperiodic.*

Under Assumption 1, for every policy π , the Markov chain (s_t) has a unique stationary distribution $d^\pi \in (0, 1)^{\mathcal{S}}$, independent of ρ , such that $(d^\pi)^\top \mathcal{P}^\pi = (d^\pi)^\top$, where for states s, s' , $\mathcal{P}^\pi(s, s') := \sum_a \pi(a|s) \mathcal{P}(s'|s, a)$.

The following is an important quantity associated with the CMDP $\mathcal{M}$.

Definition 1 (Mixing Time). *The mixing time τ_{mix}^π of $\mathcal{M}$ under policy π is defined as*

$$\tau_{\text{mix}}^\pi := \min \left\{ t > 0 : \max_s \left\| (\mathcal{P}^\pi)^t(s, \cdot) - d^\pi \right\|_{\text{TV}} \leq 1/4 \right\},$$

where $\|\cdot\|_{\text{TV}}$ denotes the total variation norm. Further, the uniform mixing time of $\mathcal{M}$ is defined as $\tau_{\text{mix}} := \sup_\pi \tau_{\text{mix}}^\pi$.

The mixing time quantifies the convergence rate of (s_t) generated by any policy π to its stationary distribution d^π . Assumption 1 implies $\tau_{\text{mix}} < \infty$ (Satheesh et al. 2026).

Given a policy π , we define its average reward J_r^π and average constraint cost J_c^π as

$$J_u^\pi := \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=0}^{T-1} u(s_t, a_t) \middle| s_0 \sim \rho \right], \quad (1)$$

¹ $\Delta(U)$ denotes the probability simplex over a finite set U .

Reference	Neural Critic	Convergence Rate	Constraint Violation	Unknown Mixing Time
Ganesh, Mondal, and Aggarwal (2025b)	✗	$\tilde{\mathcal{O}}(T^{-1/2})$	N/A	✓
Xu et al. (2025)	✗	$\tilde{\mathcal{O}}(T^{-1/2+\varepsilon})$	$\tilde{\mathcal{O}}(T^{-1/2+\varepsilon})$	✓
Xu et al. (2025)	✗	$\tilde{\mathcal{O}}(T^{-1/2})$	$\tilde{\mathcal{O}}(T^{-1/2})$	✗
Ganesh et al. (2025)	✓	$\tilde{\mathcal{O}}(T^{-1/2})$	N/A	✗
Satheesh et al. (2026)	✓	$\tilde{\mathcal{O}}(T^{-1/4})$	$\tilde{\mathcal{O}}(T^{-1/4})$	✓
Our Work	✓	$\tilde{\mathcal{O}}(T^{-1/2})$	$\tilde{\mathcal{O}}(T^{-1/2})$	✓

Table 1: Comparison with related works under general policy parameterizations. The table indicates the type of critic employed (✗ for linear vs ✓ for neural), the convergence and constraint-violation rates achieved (with “N/A” denoting unconstrained MDPs), and whether knowledge of the mixing time is required.

for each $u \in \{r, c\}$, where the expectation is taken over trajectories $(s_t, a_t)_{t=0}^\infty$ generated by policy π . Under the ergodicity assumption, we have $J_u^\pi = \mathbb{E}_{(s,a) \sim \nu^\pi} [u(s, a)]$, where ν^π denotes the stationary occupancy measure defined as $\nu^\pi(s, a) := d^\pi(s) \pi(a|s)$, for each state-action pair (s, a) .

A common practice, when dealing with a large or continuous state space, is to parameterize the policy as π_θ , where $\theta \in \Theta \subset \mathbb{R}^d$ and $d \ll |S||A|$. Hereafter, we adopt the shorthand $J_u^\theta = J_u^{\pi_\theta}$ and $\nu^\theta = \nu^{\pi_\theta}$. Further, we let $\mathbb{E}_\theta[\cdot]$ denote the expectation w.r.t. (s, a, s', a') satisfying $(s, a) \sim \nu^\theta$, $s' \sim \mathcal{P}(\cdot|s, a)$, and $a' \sim \pi_\theta(\cdot|s')$. Given these quantities, our objective is to solve the following constrained optimization:

$$\max_{\theta \in \Theta} J_r^\theta \quad \text{s.t.} \quad J_c^\theta \geq 0. \quad (2)$$

To ensure that (2) admits a feasible solution, we make the following assumption (Wei, Liu, and Ying 2022; Bai, Mondal, and Aggarwal 2024a).

Assumption 2 (Slater Condition). *There exists $\delta \in (0, 1)$ and $\bar{\theta} \in \Theta$ such that $J_c^{\bar{\theta}} \geq \delta$.*

2.1 Primal-Dual Natural Policy Gradient

The primal-dual approach reformulates the constrained optimization (2) as the saddle-point problem

$$\max_{\theta \in \Theta} \min_{\lambda \geq 0} \mathcal{L}(\theta, \lambda) := J_r^\theta + \lambda J_c^\theta, \quad (3)$$

where $\mathcal{L}$ is called the Lagrangian, and λ is referred to as the dual variable. A common gradient-based scheme for (3) is Natural Policy Gradient (NPG), defined by the updates

$$\begin{aligned} \theta_{k+1} &= \theta_k + \alpha F(\theta_k)^\dagger \nabla_\theta \mathcal{L}(\theta_k, \lambda_k), \\ \lambda_{k+1} &= \Pi_{[0, 2/\delta]} (\lambda_k - \beta J_c^{\theta_k}). \end{aligned} \quad (4)$$

Here, $F(\theta)$ is the Fisher information matrix defined as

$$F(\theta) := \mathbb{E}_\theta \left[\nabla_\theta \ln \pi_\theta(a|s) (\nabla_\theta \ln \pi_\theta(a|s))^\top \right], \quad (5)$$

$\dagger$ denotes the Moore-Penrose pseudoinverse, α and β are the primal and dual stepsizes, respectively, δ is the constant from Assumption 2, and Π_A denotes projection onto a set $A \subseteq \mathbb{R}$.

The NPG vector $w_k^* := F(\theta_k)^\dagger \nabla_\theta \mathcal{L}(\theta_k, \lambda_k)$ is obtained by minimizing the function

$$f(\theta_k, \lambda_k, w) := \frac{1}{2} w^\top F(\theta_k) w - w^\top \nabla_\theta \mathcal{L}(\theta_k, \lambda_k). \quad (6)$$

This is achieved by applying gradient descent with

$$\nabla_w f(\theta_k, \lambda_k, w) = F(\theta_k) w - (\nabla_\theta J_r^{\theta_k} + \lambda_k \nabla_\theta J_c^{\theta_k}), \quad (7)$$

which in turn requires computing $\nabla_\theta J_u^\theta$, $u \in \{r, c\}$. By the policy gradient theorem (Sutton et al. 1999), for each $u \in \{r, c\}$, we have

$$\nabla_\theta J_u^\theta = \mathbb{E}_\theta [A_u^\theta(s, a) \nabla_\theta \ln \pi_\theta(a|s)], \quad (8)$$

where $A_u^\theta \in \mathbb{R}^{|S||A|}$ denotes the advantage function defined as

$$A_u^\theta(s, a) := Q_u^\theta(s, a) - \mathbb{E}_{a' \sim \pi_\theta(\cdot|s)} [Q_u^\theta(s, a')], \quad (9)$$

and $Q_u^\theta \in \mathbb{R}^{|S||A|}$ is a solution to the Bellman equation

$$Q(s, a) = u(s, a) - J_u^\theta + \mathbb{E}_\theta [Q(s', a')]. \quad (10)$$

Such a function is unique (up to additive constants) and is called the state-action value function for the policy π_θ .

Consequently, implementing (4) hinges on estimating $(J_u^{\theta_k}, Q_u^{\theta_k})$ for $u \in \{r, c\}$. We now describe the critic estimator used for this purpose.

2.2 Neural Critic Estimation

Given a policy π_θ , the average reward/cost J_u^θ can be estimated with relative ease. However, estimating the function Q_u^θ is often computationally challenging. In this work, we employ a neural critic for this task. Let $\phi : S \times A \rightarrow \mathbb{R}^d$ be a feature map such that $\|\phi(s, a)\| \leq 1$, for every state-action pair (s, a) . Given matrices $W_u^{(1)} \in \mathbb{R}^{m \times d}$ and $\{W_u^{(\ell)}\}_{\ell=2}^L \subset \mathbb{R}^{m \times m}$, we let the vector

$$\zeta_u := \left[\text{Vec}^\top(W_u^{(1)}), \dots, \text{Vec}^\top(W_u^{(L)}) \right]^\top$$

serve as our critic parameter.² These matrices consist of trainable weights in an L -layer feedforward neural network defined as

$$x_u^{(\ell)}(s, a) := \frac{1}{\sqrt{m}} \sigma \left(W_u^{(\ell)} x_u^{(\ell-1)}(s, a) \right), \quad \ell \in [L], \quad (11)$$

where $x_u^{(0)}(s, a) := \phi(s, a)$ and σ is an element-wise activation function. For each $u \in \{r, c\}$, the output of this neural network lets us approximate $Q_u^\theta(s, a)$ as

$$Q(s, a, \zeta_u) := \frac{1}{\sqrt{m}} b_u^\top x_u^{(L)}(s, a), \quad (12)$$

²Vec($\cdot$) denotes vectorization by stacking columns.

where $b_u \in \mathbb{R}^m$ is a fixed vector.

Throughout the paper, we make the following assumption.

Assumption 3 (Smooth Activation). *The activation function σ is L_1 -Lipschitz and L_2 -smooth.*

Remark 1. *Assumption 3 is satisfied by commonly used smooth approximations of ReLU such as Sigmoid, ELU, and GeLU (Hendrycks and Gimpel 2016; Clevert, Unterthiner, and Hochreiter 2015).*

For subsequent analysis, we rely on the Neural Tangent Kernel (NTK) regime, which requires restricting the critic parameter ζ_u to a ball $\mathcal{B}$ of radius R around the initialization ζ_0 . This lets us approximate the neural network output $Q(\cdot, \cdot, \zeta)$ by the linear function class

$$\mathcal{F}_{\mathcal{B},m} := \left\{ Q_{\text{lin}}(\cdot, \cdot, \zeta) : \zeta \in \mathcal{B} \right\}, \quad (13)$$

where $Q_{\text{lin}}(\cdot, \cdot, \zeta_0)$ is defined as

$$Q_{\text{lin}}(\cdot, \cdot, \zeta) := Q(\cdot, \cdot, \zeta_0) + (\zeta - \zeta_0)^\top \nabla_\zeta Q(\cdot, \cdot, \zeta_0). \quad (14)$$

3 Algorithm

We now present our hierarchical MLMC-based primal-dual Natural Actor-Critic algorithm (HiMLMC-PD-NAC) for solving (2), with its pseudocode provided in Algorithm 1. As the transition kernel and occupancy measures are unknown, we rely on sample-based estimates.

Our algorithm updates the primal-dual pair (θ_k, λ_k) via the update rule

$$\begin{aligned} \theta_{k+1} &= \theta_k + \alpha w_k \\ \lambda_{k+1} &= \Pi_{[0,2/\delta]}(\lambda_k - \beta \eta_{c,k}), \end{aligned} \quad (15)$$

where α, β , and δ are as in (4), whereas w_k and $\eta_{c,k}$ are approximations to the NPG vector w_k^* and the average-constraint-cost $J_c^{\theta_k}$ associated with policy π_{θ_k} , respectively. The update (15) is run for $K > 0$ iterations, yielding the output $\{(\theta_k, \lambda_k)\}_{k < K}$.

3.1 Critic Estimator

For each outer-loop iteration $0 \leq k < K$, we approximate the critic $(J_u^{\theta_k}, Q_u^{\theta_k})$ by an estimator $(\eta_{u,k,\text{MLMC}}, Q_{u,k,\text{MLMC}})$. The critic bias is controlled using a hierarchical two-layer MLMC scheme with outer- and inner-layer truncation parameters $H_{\max}$ and $T_{\max}$, respectively. The outer MLMC layer operates on several outputs from a vanilla-critic subroutine, with each vanilla output corresponding to a different number of iterations. Specifically, after h_k iterations, the vanilla subroutine produces the output ξ_{u,h_k}^k . These vanilla outputs are then combined by the outer layer to form $(\eta_{u,k,\text{MLMC}}, Q_{u,k,\text{MLMC}})$. The inner MLMC layer is used for gradient estimation within the vanilla subroutine.

Outer MLMC layer: Sample $H_k \sim \text{Geom}(1/2)$ and let $h_k := 1 + \mathbb{1}_{\{2^{H_k} \leq H_{\max}\}}(2^{H_k} - 1)$. Then, we define the critic estimation function

$$\begin{aligned} Q_{u,k,\text{MLMC}}(\cdot, \cdot) &:= Q(\cdot, \cdot, \zeta_{u,1}^k) + \mathbb{1}_{\{2^{H_k} \leq H_{\max}\}} \\ &\times 2^{H_k} \left[Q(\cdot, \cdot, \zeta_{u,h_k}^k) - Q(\cdot, \cdot, \zeta_{u,h_k/2}^k) \right] \end{aligned} \quad (16)$$

and the MLMC critic parameter

$$\begin{aligned} \xi_{u,k,\text{MLMC}} &= [\eta_{u,k,\text{MLMC}}, (\zeta_{u,k,\text{MLMC}})^\top]^\top \\ &:= \xi_{u,1}^k + \mathbb{1}_{\{2^{H_k} \leq H_{\max}\}} 2^{H_k} \left[\xi_{u,h_k}^k - \xi_{u,h_k/2}^k \right], \end{aligned} \quad (17)$$

where for $H \in \{1, h_k, h_k/2\}$, $\xi_{u,H}^k$ denotes the output from the vanilla-critic subroutine after H iterations.

Inner MLMC layer: The vanilla-critic subroutine seeks to solve the optimization

$$\begin{aligned} \min_{\eta} \mathcal{R}_u(\theta_k, \eta) &:= \frac{1}{2} \mathbb{E}_{\theta_k} \left[(\eta - u(s, a))^2 \right] \\ \min_{\zeta} \mathcal{E}_u(\theta_k, \zeta) &:= \frac{1}{2} \mathbb{E}_{\theta_k} \left[(Q(s, a, \zeta) - Q_u^{\theta_k}(s, a))^2 \right]. \end{aligned}$$

We optimize these objectives together using Stochastic Gradient Descent (SGD). At each iteration $0 \leq h < h_k$, we estimate the joint gradient $(\nabla_{\eta} \mathcal{R}_u(\theta_k, \eta_{u,h}^k), \nabla_{\zeta} \mathcal{E}_u(\theta_k, \zeta_{u,h}^k))$ and update the vector

$$\xi_{u,h}^k = [\eta_{u,h}^k, (\zeta_{u,h}^k)^\top]^\top.$$

To reduce gradient bias, we employ a second MLMC construction. Let $U_{k,h} \sim \text{Geom}(1/2)$ and define

$$\ell_{k,h} := 1 + \mathbb{1}_{\{2^{U_{k,h}} \leq T_{\max}\}} (2^{U_{k,h}} - 1). \quad (18)$$

Given a trajectory $\mathcal{T}^{k,h} := (s_t^{k,h}, a_t^{k,h})_{t=0}^{\ell_{k,h}}$ generated by the policy π_{θ_k} , we define the naive gradient-estimator as $\hat{v}_{u,T}^{k,h} := \frac{1}{T} \sum_{t=0}^{T-1} \hat{v}_u^{k,h}(s_t^{k,h}, a_t^{k,h}, s_{t+1}^{k,h}, a_{t+1}^{k,h})$, with

$$\hat{v}_u^{k,h}(s, a, s', a') := \left[\hat{\nabla}_{\zeta} \hat{\nabla}_{\eta} \mathcal{R}_u^{k,h}(s, a) \right], \quad (19)$$

where $c_\gamma > 0$ is a scaling parameter, and

$$\begin{aligned} \hat{\nabla}_{\eta} \mathcal{R}_u^{k,h}(s, a) &:= \eta_{u,h}^k - u(s, a), \\ \hat{\nabla}_{\zeta} \mathcal{E}_u^{k,h}(s, a, s', a') &:= \nabla_{\zeta} Q(s, a, \zeta_0) \\ &\times (\eta_{u,h}^k - u(s, a) + Q(s, a, \zeta_{u,h}^k) - Q(s', a', \zeta_{u,h}^k)). \end{aligned}$$

Then, the corresponding MLMC-based gradient estimator is defined as

$$\begin{aligned} \hat{v}_{u,\text{MLMC}}^{k,h} &:= \hat{v}_{u,1}^{k,h} + \mathbb{1}_{\{2^{U_{k,h}} \leq T_{\max}\}} \\ &\times 2^{U_{k,h}} \left(\hat{v}_{u,2^{U_{k,h}}}^{k,h} - \hat{v}_{u,2^{U_{k,h}-1}}^{k,h} \right). \end{aligned} \quad (20)$$

Finally, we run the following update rule for H iterations:

$$\xi_{u,h+1}^k = \Pi_{\mathcal{B}} \left(\xi_{u,h}^k - \gamma_{\xi,H} \hat{v}_{u,\text{MLMC}}^{k,h} \right), \quad (21)$$

where $\gamma_{\xi,H}$ is a stepsize dependent on H , while $\Pi_{\mathcal{B}}$ projects only the ζ portion onto the NTK ball $\mathcal{B}$. After completing H iterations, with each $H \in \{1, h_k/2, h_k\}$, we have the quantities required in (17).

3.2 NPG Estimator

For each outer iteration $0 \leq k < K$, this subroutine estimates the NPG vector w_k^* associated with the current policy π_{θ_k} . As discussed in Section 2.1, the NPG vector is obtained by minimizing $f(\theta_k, \lambda_k, w)$ defined in (6). To employ an MLMC-based SGD scheme, we sample $U_{k,h} \sim \text{Geom}(1/2)$, and generate a trajectory $\mathcal{T}^{k,h} := (s_t^{k,h}, a_t^{k,h})_{t=0}^{\ell_{k,h}}$ using policy π_{θ_k} , where $\ell_{k,h}$ is as defined in (18). We define the naive T -sample estimators for $F(\theta_k)$ and $\nabla_{\theta} J_u^{\theta_k}$ as

$$\hat{F}_T^{k,h} := \frac{1}{T} \sum_{t=0}^{T-1} \hat{F}_t^{k,h}(s_t^{k,h}, a_t^{k,h}),$$

$$\text{and } \hat{g}_{u,T}^{k,h} := \frac{1}{T} \sum_{t=0}^{T-1} \hat{g}_u^{k,h}(s_t^{k,h}, a_t^{k,h}, \bar{a}_t^{k,h}, s_{t+1}^{k,h}, a_{t+1}^{k,h}),$$

respectively, where $\bar{a}_t^{k,h} \sim \pi_{\theta_k}(\cdot | s_t^{k,h})$ and

$$\begin{aligned} \hat{F}_t^{k,h}(s, a) &:= \nabla_{\theta} \ln \pi_{\theta_k}(a|s) (\nabla_{\theta} \ln \pi_{\theta_k}(a|s))^{\top}, \\ \hat{g}_u^{k,h}(s, a, \bar{a}, s', a') &:= \nabla_{\theta} \ln \pi_{\theta_k}(a|s) (u(s, a) - \eta_{u,k} \\ &\quad + Q_{u,k,\text{MLMC}}(s', a') - Q_{u,k,\text{MLMC}}(s, \bar{a})). \end{aligned} \quad (22)$$

Then, the corresponding MLMC estimators are defined as

$$\begin{aligned} \hat{X}_{\text{MLMC}}^{k,h} &:= \hat{X}_1^{k,h} + \mathbb{1}_{\{2^{U_{k,h}} \leq T_{\max}\}} \\ &\quad \times 2^{U_{k,h}} (\hat{X}_{2^{U_{k,h}}}^{k,h} - \hat{X}_{2^{U_{k,h}-1}}^{k,h}), \end{aligned} \quad (23)$$

for each $\hat{X}^{k,h} \in \{\hat{F}^{k,h}, \hat{g}_u^{k,h}\}$. Finally, with these estimates, we run the following SGD update

$$\begin{aligned} w_{h+1}^k &= w_h^k \\ &\quad - \gamma_w \left[\hat{F}_{\text{MLMC}}^{k,h} w_h^k - (\hat{g}_{r,\text{MLMC}}^{k,h} + \lambda_k \hat{g}_{c,\text{MLMC}}^{k,h}) \right]. \end{aligned} \quad (24)$$

After H_w iterations, we set $w_k := w_{H_w}^k$ as the NPG estimate for policy π_{θ_k} .

Remark 2 (Expected Sample Cost). Since H_k and $U_{k,h}$ are independent, we have $\mathbb{E}[h_k \ell_{k,h}] = \mathbb{E}[h_k] \cdot \mathbb{E}[\ell_{k,h}] = \mathcal{O}(\lceil \log_2 H_{\max} \rceil \cdot \lceil \log_2 T_{\max} \rceil)$. Thus, for each outer iteration, our expected critic-estimation cost is $\mathcal{O}(\ln H_{\max} \cdot \ln T_{\max})$. Similarly, our NPG-estimation cost is $\mathcal{O}(H_w \cdot \ln T_{\max})$.

4 Assumptions and Main Results

Before we present our finite-time convergence bounds in Theorem 1, we state the following assumptions.

Assumption 4 (Score Function). There exist $G_1, G_2 > 0$, such that for all $\theta, \theta' \in \Theta$, and state-action pair (s, a) ,

$$\begin{aligned} \|\nabla_{\theta} \ln \pi_{\theta}(a|s)\| &\leq G_1 \quad \text{and} \\ \|\nabla_{\theta} \ln \pi_{\theta}(a|s) - \nabla_{\theta'} \ln \pi_{\theta'}(a|s)\| &\leq G_2 \|\theta - \theta'\|. \end{aligned}$$

Assumption 5 (Fisher Non-Degeneracy). There exists $\mu_F > 0$, such that for all $\theta \in \Theta$, $F(\theta) \succeq \mu_F I$.

Assumption 6 (Policy Parametrization Error). There exists $\varepsilon_{\text{bias}} > 0$ such that for all $\theta \in \Theta$ and $\lambda \in [0, 2/\delta]$, we have

$$\min_{w \in \mathbb{R}^d} L_{\nu^{\pi^*}}(\theta, \lambda, w) \leq \varepsilon_{\text{bias}},$$

where π^* and ν^{π^*} denote an optimal stationary policy³ and its occupancy measure, respectively, while the function $L_{\nu}(\theta, \lambda, w)$ is called the transferred compatible function and is defined as

$$\mathbb{E}_{\nu} \left[(w^{\top} \nabla_{\theta} \ln \pi_{\theta}(a|s) - A_r^{\theta}(s, a) - \lambda A_c^{\theta}(s, a))^2 \right].$$

Remark 3. Assumptions 4–6 are standard in the PG literature (Liu et al. 2020; Papini et al. 2018; Xu, Gao, and Gu 2020; Fatkhullin et al. 2023; Ding et al. 2025). Assumption 4 requires the score function to be bounded and Lipschitz, and is satisfied by Gaussian policies with bounded variance and softmax policies. Assumption 5 ensures that $f(\theta, \lambda, w)$ is strongly convex and admits a unique minimizer. Assumption 6 characterizes the richness of the policy class.

Assumption 7 (Critic Approximation Error). There exists a finite $\varepsilon_{\text{app}} > 0$, such that for every $u \in \{r, c\}$, and $\theta \in \Theta$, we have

$$\mathbb{E}_{\theta} \left[(Q_u^{\theta}(s, a) - (\Pi_{\mathcal{F}_{\mathcal{B},m}} Q_u^{\theta})(s, a))^2 \right] \leq \varepsilon_{\text{app}},$$

where $\Pi_{\mathcal{F}_{\mathcal{B},m}}$ denotes projection onto the linear function class defined in (13).

Remark 4. Assumption 7 ensures that the linear function class $\mathcal{F}_{\mathcal{B},m}$ approximates the state-action value function with bounded error. Under NTK, it is equivalent to (Gaur et al. 2024, Assumption 2.5) and (Cayci, He, and Srikant 2024, Assumption 2).

We make one final assumption commonly used for average-reward MDPs (Ganesh, Mondal, and Aggarwal 2025b; Xu et al. 2025; Satheesh et al. 2026).

Assumption 8 (Smooth Objectives). The average-reward J_r^{θ} and the average-constraint-cost J_c^{θ} are L -smooth in the policy parameter θ .

Assumption 8 is satisfied by several commonly used policy parameterizations, including the softmax policy. Under Assumptions 1–8, we have the following result.

Theorem 1 (Global Convergence and Constraint Violation). Choose the vanilla-critic stepsizes $\gamma_{\xi, H} = \frac{2 \ln T}{\mu_{\phi} H}$, with suitable $\mu_{\phi} > 0$, and the NPG stepsize $\gamma_w = \frac{2 \ln T}{\mu_F H_w}$. Set the network width $m = H_{\max}$, and the NTK radius $R = \Theta(\ln T)$. Set the primal-dual stepsizes $\alpha = \beta = \frac{1}{\sqrt{K}}$, and horizon lengths $T_{\max} = H_{\max} = K = \Theta(T)$, and $H_w = \Theta(\ln T)$. Then, we have

$$\begin{aligned} J_r^{\pi^*} - \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [J_r^{\theta_k}] &= \tilde{\mathcal{O}} \left(\frac{1}{\sqrt{T}} + \sqrt{\varepsilon_{\text{bias}}} + \sqrt{\varepsilon_{\text{app}}} \right), \\ - \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [J_c^{\theta_k}] &= \tilde{\mathcal{O}} \left(\frac{1}{\sqrt{T}} + \sqrt{\varepsilon_{\text{bias}}} + \sqrt{\varepsilon_{\text{app}}} \right), \end{aligned}$$

where π^* is as defined in Assumption 6.

³ π^* is a solution to the constrained optimization: $\max_{\pi} J_r^{\pi}$ subject to $J_c^{\pi} \geq 0$.

Remark 5 (Optimal Global Convergence). *Theorem 1 shows that, under the prescribed parameter choices, the optimality gap decays to zero at a rate of $\tilde{O}(1/\sqrt{T})$, modulo the policy parametrization error $\varepsilon_{\text{bias}}$ and the neural function approximation error ε_{app} . Since the total expected sample cost is $K(H_w + \ln H_{\max}) \ln T_{\max} = \tilde{O}(T)$, our convergence rate is optimal.*

Remark 6 (Constraint Violation Rate). *The expected constraint violation rate also decays to zero at the rate of $\tilde{O}(1/\sqrt{T})$ (up to the parametrization and neural critic approximation errors), matching the existing state-of-the-art (Xu et al. 2025).*

Remark 7 (Mixing-Time Knowledge). *The parameter choices in Theorem 1 require no knowledge of the mixing time of the underlying CMDP.*

5 Proof Sketch

We now outline the proof of Theorem 1. Our proof consists of the following key steps.

1. Use the dual update in (15) to express the **optimality gap and constraint violation** as a combination of the Lagrangian error and critic bounds.
2. Bound the **Lagrangian error** using the primal update in (15) in terms of the NPG bias and MSE.
3. Exploit the linear structure of the NPG update (24) together with MLMC bounds for the Fisher matrix and policy-gradient estimates to control the **NPG bias and MSE** in terms of the critic parameter.
4. Bound the critic bias and MSE:
 - (a) Bound the **vanilla critic**: decompose the vanilla critic update (21) into a perturbed linear recursion, then control the linear components using the inner MLMC layer (20), and the nonlinear perturbation via the NTK-regime.
 - (b) Bound the **MLMC critic**: combine the vanilla-critic bounds with the outer MLMC layer (17).

Before we elaborate on these steps, we introduce the following notation: let $\mathbb{E}_k[\cdot] := \mathbb{E}[\cdot | \mathcal{F}_k]$, where $\mathcal{F}_k$ denotes the σ -algebra generated by all samples till the $(k-1)$ -th outer loop iteration.

Step 1. Optimality Gap and Constraint Violation: We borrow from the proof strategy of (Ganesh, Mondal, and Aggarwal 2025b), later adapted for primal-dual methods with a linear critic by (Xu et al. 2025). Let $\mathcal{L}(\pi^*, \lambda)$ be defined as $J_r^{\pi^*} + \lambda J_c^{\pi^*}$, where π^* is an optimal stationary policy. Using the dual update in (15) and the definition of $\mathcal{L}$, we can bound the optimality gap and constraint violation as given by the following two lemmas.

Lemma 1 (Optimality Gap). *Under Assumption 2, we have*

$$\begin{aligned} & \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}[J_r^{\pi^*} - J_r^{\theta_k}] \\ &= \mathcal{O} \left(\frac{1}{K} \sum_{k=0}^{K-1} \left[\beta \mathbb{E}_k \|\xi_{c,k}^* - \xi_{c,k,\text{MLMC}}\|^2 \right. \right. \\ & \quad \left. \left. + \mathbb{E} \|\mathbb{E}_k [\xi_{c,k,\text{MLMC}}] - \xi_{c,k}^*\| \right] \right. \\ & \quad \left. + \beta + \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\mathcal{L}(\pi^*, \lambda_k) - \mathcal{L}(\theta_k, \lambda_k)] \right). \end{aligned}$$

Lemma 2 (Constraint Violation Rate). *Under Assumption 2, we have*

$$\begin{aligned} -\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}[J_c^{\theta_k}] &= \mathcal{O} \left(\frac{1}{K} \sum_{k=0}^{K-1} \left[\beta \mathbb{E} \|\xi_{c,k,\text{MLMC}} - \xi_{c,k}^*\|^2 \right. \right. \\ & \quad \left. \left. + \mathbb{E} \|\mathbb{E}_k [\xi_{c,k,\text{MLMC}}] - \xi_{c,k}^*\| \right] \right. \\ & \quad \left. + \beta + \frac{1}{\beta K} + \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\mathcal{L}(\pi^*, \lambda_k) - \mathcal{L}(\theta_k, \lambda_k)] \right). \end{aligned}$$

The detailed proofs of Lemmas 1 and 2 are deferred to Appendices A and B, respectively.

Step 2. Lagrangian Error: In light of Lemmas 1 and 2, our goal for the remainder of this section is to bound the Lagrangian error. Under Assumption 8, we have the following result.

Lemma 3. *Let (θ_k, λ_k) be updated by (15). Under Assumptions 4–6, the following holds for $0 \leq k < K$:*

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\mathcal{L}(\pi^*, \lambda_k) - \mathcal{L}(\theta_k, \lambda_k)] &= \mathcal{O} \left(\sqrt{\varepsilon_{\text{bias}}} + \alpha + \frac{1}{\alpha K} \right. \\ & \quad \left. + \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\mathbb{E}_k[w_k] - w_k^*\| + \frac{\alpha}{K} \sum_{k=0}^{K-1} \mathbb{E} \|w_k - w_k^*\|^2 \right). \end{aligned}$$

The proof of Lemma 3 is given in Appendix C.

Step 3. NPG Estimator Analysis: The NPG update (24) is a stochastic linear recursion with the mean drift $(\nabla_{\theta} J_r^{\theta_k} + \lambda_k \nabla_{\theta} J_c^{\theta_k}) - F(\theta_k)w$. Therefore, standard arguments on stochastic linear recursions allow us to bound the NPG bias and MSE in terms of the biases and MSEs of the Fisher and policy gradient estimates. For the Fisher matrix, the bias arises from replacing the occupancy measure ν^{θ_k} with finite-length trajectories generated by π_{θ_k} . Standard MLMC arguments yield a squared bias of order $\mathcal{O}(\tau_{\text{mix}}/T_{\max})$ and MSE of order $\mathcal{O}(\ln T_{\max})$. In contrast, the advantage function (9) contributes additional bias to the PG estimator $\hat{g}_{u,\text{MLMC}}^{k,h}$ through the neural critic approximation. The NTK regime lets us replace the neural critic $Q(\cdot, \cdot, \zeta)$ by its first-order Taylor approximation $Q_{\text{lin}}(\cdot, \cdot, \zeta) \in \mathcal{F}_{\mathcal{B},m}$. Let $\xi_{u,k}^* := [\eta_{u,k}^*, (\zeta_{u,k}^*)^\top]^\top$, where $\eta_{u,k}^* := J_u^{\theta_k}$ and $\zeta_{u,k}^*$ is a

minimum-norm solution to the following projected Bellman equation

$$\mathbb{E}_{\theta_k} \left[\nabla_{\zeta} Q(s, a, \zeta_0) (J_{u,k}^{\theta_k} - u(s, a) + Q_{\text{lin}}(s, a, \zeta) - Q_{\text{lin}}(s', a', \zeta)) \right] = 0. \quad (25)$$

Then, we have the following result.

Theorem 2 (NPG bounds). *Let Assumptions 1–7 hold and choose $\gamma_w = \frac{2 \ln T}{\mu_F H_w}$. Then, the following bounds hold:*

$$\begin{aligned} \|\mathbb{E}_k[w_k] - w_k^*\|^2 &= \tilde{\mathcal{O}} \left(\frac{1}{T^2} + \frac{\tau_{\text{mix}}}{T_{\text{max}}} + \frac{1}{m} + \varepsilon_{\text{app}} \right. \\ &\quad \left. + \delta_Q^2 + \sum_{u \in \{r, c\}} \|\mathbb{E}_k[\xi_{u,k,\text{MLMC}}] - \xi_{u,k}^*\|^2 \right), \\ \mathbb{E}_k \|w_k - w_k^*\|^2 &= \tilde{\mathcal{O}} \left(\frac{1}{T^2} + \tau_{\text{mix}} + \frac{1}{m} + \varepsilon_{\text{app}} \right. \\ &\quad \left. + \sigma_Q^2 + \sum_{u \in \{r, c\}} \mathbb{E}_k \|\xi_{u,k,\text{MLMC}} - \xi_{u,k}^*\|^2 \right), \end{aligned}$$

where

$$\begin{aligned} \delta_Q^2 &:= \sum_u \|\mathbb{E}_k[Q_{u,k,\text{MLMC}} - Q_{\text{lin}}(\cdot, \cdot, \zeta_{u,k,\text{MLMC}})]\|^2, \\ \sigma_Q^2 &:= \sum_u \mathbb{E}_k \|Q_{u,k,\text{MLMC}} - Q_{\text{lin}}(\cdot, \cdot, \zeta_{u,k,\text{MLMC}})\|^2. \end{aligned}$$

Inside the NTK ball, the neural critic $Q(\cdot, \cdot, \zeta)$ decomposes into the linear critic $Q_{\text{lin}}(\cdot, \cdot, \zeta)$ and a nonlinear remainder. This nonlinear remainder yields the $\tilde{\mathcal{O}}(1/m)$ term in Theorem 2. Whereas, ε_{app} reflects the approximation capacity of the linear critic.

Consequently, the analysis is reduced to controlling the critic bias and MSE. Deferring the proof of Theorem 1 to Appendix D, we now establish the critic bounds in the following step.

Step 4. Critic Estimator Analysis: Recall the hierarchical MLMC scheme introduced in Section 3.1. Since the outer layer combines outputs of the vanilla-critic subroutine corresponding to different iteration budgets, as in (17), we begin by analyzing the vanilla-critic $\xi_{u,H}^k$ generated by (21) with an arbitrary iteration budget H .

The NTK projection $\Pi_{\mathcal{B}}$ introduces an additional error in the bias analysis of $\xi_{u,H}^k$. This projection error is controlled by the probability with which iterates escape the NTK ball, which, in turn, is bounded in terms of the vanilla-critic MSE (see (Ganesh et al. 2025, Appendix A.2)). To bound the vanilla-critic MSE, we decompose (21) as a perturbed linear recursion. Under the NTK regime, the linear critic $Q_{\text{lin}}(\cdot, \cdot, \zeta)$ approximates the neural critic up to an error $\tilde{\mathcal{O}}(m^{-1/2})$ (see Lemma 6). By leveraging this linear approximation, we can write (21) as:

$$\begin{aligned} \xi_{u,h+1}^k &= \Pi_{\mathcal{B}} \left(\xi_{u,h}^k + \gamma_{\xi,H} [\hat{B}_{u,\text{MLMC}}^{k,h} - \hat{A}_{\text{MLMC}}^{k,h} \xi_{u,h}^k] \right. \\ &\quad \left. + \gamma_{\xi,H} \tilde{V}_{u,\text{MLMC}}^{k,h} \right), \end{aligned}$$

where the linear component $(\hat{B}_{u,\text{MLMC}}^{k,h} - \hat{A}_{\text{MLMC}}^{k,h} \xi)$ is obtained by replacing $Q(\cdot, \cdot, \zeta)$ with $Q_{\text{lin}}(\cdot, \cdot, \zeta)$ in (19) and

using the inner-layer MLMC construction defined in (20). Collecting the remainder yields the nonlinear component $\tilde{V}_{u,\text{MLMC}}^{k,h}$. Under the NTK regime, we have $\|\tilde{V}_{u,\text{MLMC}}^{k,h}\| = \tilde{\mathcal{O}}(m^{-1/2})$, while standard MLMC bounds apply on the linear component. This results in the following bound for the vanilla critic (detailed proof in Appendix E).

Lemma 4 (Vanilla-Critic Bounds). *Let $\xi_{u,H}^k$ be generated by (21) after H iterations and let Assumptions 1–7 hold. Choose $\gamma_{\xi,H} = \frac{2 \ln T}{\mu_{\phi} H}$, for sufficiently small μ_{ϕ} , and let c_{γ} be sufficiently large. Then, for all $u \in \{r, c\}$, we have*

$$\mathbb{E}_k \|\xi_{u,H}^k - \xi_{u,k}^*\|^2 = \tilde{\mathcal{O}} \left(\frac{1}{T^2} + \frac{\tau_{\text{mix}}}{T_{\text{max}}} + \frac{\tau_{\text{mix}}}{H} + \frac{\tau_{\text{mix}}}{m} \right).$$

Remark 8 (Bias-Cost Tradeoff). *Owing to the NTK projection $\Pi_{\mathcal{B}}$, our squared critic bias has the same $\tilde{\mathcal{O}}(H^{-1})$ rate as the MSE. In contrast, a linear critic yields squared bias of order $\tilde{\mathcal{O}}(T_{\text{max}}^{-1})$ (Xu et al. 2025, Lemma 4.8). However, our primal-dual scheme requires a large outer-loop size K to control constraint violations, making a large H prohibitively expensive. Hence, directly using $\xi_{u,H}^k$ for gradient estimation introduces a fundamental bias-cost tradeoff.*

We overcome the above bias-cost tradeoff by exploiting the outer MLMC layer defined by (16) and (17). Recall that the iteration budget for the vanilla subroutine is set to $h_k := 1 + \mathbb{1}_{\{2^{H_k} \leq H_{\text{max}}\}} (2^{H_k} - 1)$ with $H_k \sim \text{Geom}(1/2)$. A telescoping-sum argument then shows that

$$\mathbb{E}[\xi_{u,\text{MLMC}}] = \mathbb{E}[\xi_{u,H_{\text{max}}}^k] \quad \text{and} \quad \mathbb{E}[h_k] = \lfloor \log_2 H_{\text{max}} \rfloor.$$

Therefore, while the MLMC estimator inherits the bias of a vanilla critic updated for H_{max} iterations, the expected number of vanilla iterations is only $\mathcal{O}(\ln H_{\text{max}})$. Using a similar argument for the MSE, we prove the following bounds in Appendix E.

Theorem 3 (MLMC-Critic Bounds). *Consider the setting of Lemma 4. Then, the critic function $Q_{u,k,\text{MLMC}}$ satisfies $\delta_Q^2 = \tilde{\mathcal{O}}(1/m)$ and $\sigma_Q^2 = \tilde{\mathcal{O}}(H_{\text{max}}/m)$. Moreover, the critic parameter $\xi_{u,k,\text{MLMC}}$ satisfies:*

$$\begin{aligned} \mathbb{E}_k \|\xi_{u,k,\text{MLMC}} - \xi_{u,k}^*\|^2 &= \tilde{\mathcal{O}}(H_{\text{max}}/m), \\ \|\mathbb{E}_k[\xi_{u,k,\text{MLMC}}] - \xi_{u,k}^*\|^2 &= \tilde{\mathcal{O}} \left(\frac{\tau_{\text{mix}}}{T_{\text{max}}} + \frac{\tau_{\text{mix}}}{H_{\text{max}}} + \frac{\tau_{\text{mix}}}{m} \right). \end{aligned}$$

Setting $m = H_{\text{max}}$ results in a critic MSE of order $\tilde{\mathcal{O}}(1)$ and a critic bias of order $\tilde{\mathcal{O}}(T_{\text{max}}^{-1/2} + H_{\text{max}}^{-1/2})$.

Finally, in Appendix F, we combine Lemmas 1–3 with Theorems 2 and 3, and apply the prescribed parameter choices to complete the proof of Theorem 1. $\square$

6 Conclusion

In this work, we study infinite-horizon average-reward CMDPs under general policy parameterization and neural critic estimation. We propose a primal-dual natural actor-critic algorithm equipped with a novel hierarchical MLMC-based critic and establish optimal convergence guarantees for neural critics in the CMDP literature.

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Algorithm 1: HiMLMC-PD-NAC

Require: $\theta_0, \lambda_0 = 0, \alpha, \beta, \gamma_w, \gamma_\xi(\cdot), K, H_w, H_{\max}, T_{\max}$
1: **for** $k = 0, \dots, K - 1$ **do**
2: **for** $u \in \{r, c\}$ **do**
3: Compute critic $(Q_{u,k}, \eta_{u,k}) \leftarrow \text{Algorithm 2}(\theta_k, u, H_{\max}, T_{\max})$
4: **end for**
5: Compute NPG estimate $w_k \leftarrow \text{Algorithm 3}(\theta_k, \lambda_k, Q_{r,k}, Q_{c,k}, H_w, T_{\max})$
6: Update primal variable $\theta_{k+1} \leftarrow \theta_k + \alpha w_k$
7: Update dual variable $\lambda_{k+1} \leftarrow \Pi_{[0, 2/\delta]}(\lambda_k - \beta \eta_{c,k})$
8: **end for**
9: **return** $\{\theta_k, \lambda_k\}_{k=0}^{K-1}$

Algorithm 2: Hierarchical MLMC Critic

Require: $\theta_k, u, H_{\max}, T_{\max}$
1: Sample $H_k \sim \text{Geom}(1/2)$
2: $h_k = 1 + (2^{H_k} - 1) \mathbf{1}_{\{2^{H_k} \leq H_{\max}\}}$
3: Initialize $\xi_{u,0}^{k,(0)}, \xi_{u,0}^{k,(1)}, \xi_{u,0}^{k,(2)}$
4: **for** $h = 0, \dots, h_k - 1$ **do**
5: Sample $U_{k,h} \sim \text{Geom}(1/2)$ and compute $\ell_{k,h}$ using (18)
6: Generate trajectory $\mathcal{T}^{k,h} := (s_t^{k,h}, a_t^{k,h})_{t=0}^{\ell_{k,h}}$ using current policy π_{θ_k} ,
7: Construct MLMC critic gradient $\hat{v}_{u,\text{MLMC}}^{k,h}$ using (20)
8: **if** $h < 1$ **then**
9: Update $\xi_{u,h}^{k,(0)}, \xi_{u,h}^{k,(1)}$, and $\xi_{u,h}^{k,(2)}$, using (21) with stepsizes $\gamma_\xi(1), \gamma_\xi(h_k/2), \gamma_\xi(h_k)$
10: **end if**
11: **if** $h < h_k/2$ **then**
12: Update $\xi_{u,h}^{k,(1)}$ and $\xi_{u,h}^{k,(2)}$ using (21) with stepsizes $\gamma_\xi(h_k/2)$ and $\gamma_\xi(h_k)$
13: **end if**
14: Update $\xi_{u,h}^{k,(2)}$ (21) with stepsize $\gamma_\xi(h_k)$
15: **end for**
16: Compute critic function $Q_{u,k,\text{MLMC}}(\cdot, \cdot)$ and parameter $\xi_{u,k,\text{MLMC}}$ using (16) and (17)
17: **return** $(Q_{u,k,\text{MLMC}}(\cdot, \cdot), \eta_{u,k,\text{MLMC}})$

Algorithm 3: MLMC NPG Estimator

Require: $\theta_k, \lambda_k, Q_{r,k}, Q_{c,k}, H_w, T_{\max}$
1: Initialize w_0^k
2: **for** $h = 0, \dots, H_w - 1$ **do**
3: Sample $U_{k,h} \sim \text{Geom}(1/2)$ and compute $\ell_{k,h}$ using (18)
4: Generate trajectory $\mathcal{T}^{k,h} := (s_t^{k,h}, a_t^{k,h})_{t=0}^{\ell_{k,h}}$ using current policy π_{θ_k} ,
5: Construct MLMC Fisher estimate $\hat{F}_{\text{MLMC}}^{k,h}$ and MLMC PG estimate $\hat{g}_{u,\text{MLMC}}^{k,h}$ using (23)
6: Update NPG estimate w_h^k using (24)
7: **end for**
8: **return** $w_{H_w}^k$

A Bounding Optimality Gap: Proof of Lemma 1

Using the fact that $\lambda_0 = 0$, we have for all $K > 0$,

$$\begin{aligned}
\lambda_K^2 &= \sum_{k=0}^{K-1} (\lambda_{k+1}^2 - \lambda_k^2) \\
&\stackrel{(a)}{=} \sum_{k=0}^{K-1} (\Pi_{[0,2/\delta]}[\lambda_k - \beta\eta_{c,k,\text{MLMC}}])^2 - (\Pi_{[0,2/\delta]}(\lambda_k))^2 \\
&\stackrel{(b)}{\leq} \sum_{k=0}^{K-1} (\lambda_k - \beta\eta_{c,k,\text{MLMC}})^2 - \lambda_k^2 \\
&\leq \beta^2 \sum_{k=0}^{K-1} \eta_{c,k,\text{MLMC}}^2 - 2\beta \sum_{k=0}^{K-1} \lambda_k \eta_{c,k,\text{MLMC}} \\
&\stackrel{(c)}{\leq} 2\beta^2 \sum_{k=0}^{K-1} |\eta_{c,k,\text{MLMC}} - J_c^{\theta_k}|^2 + 2\beta^2 \sum_{k=0}^{K-1} (J_c^{\theta_k})^2 \\
&\quad + 2\beta \sum_{k=0}^{K-1} \lambda_k (J_c^{\theta_k} - \eta_{c,k,\text{MLMC}}) \\
&\quad + 2\beta \sum_{k=0}^{K-1} \lambda_k (J_c^{\pi^*} - J_c^{\theta_k}),
\end{aligned}$$

where (a) uses from the dual update in (15), (b) follows since any projection is non-expansive, and (c) follows since $J_c^{\pi^*} \geq 0$, and the fact that $c(s, a) \leq 1$, for all s, a implies $\eta_{c,k} \leq 1$. Since $\lambda_K \geq 0$, dividing both sides by $2\beta K$ and taking conditional expectation gives us

$$\begin{aligned}
&-\frac{1}{K} \sum_{k=0}^{K-1} \lambda_k [J_c^{\pi^*} - J_c^{\theta_k}] \\
&\leq \beta + \frac{1}{K} \sum_{k=0}^{K-1} \lambda_k (J_c^{\theta_k} - \mathbb{E}_k[\eta_{c,k,\text{MLMC}}]) \\
&\quad + \frac{\beta}{2K} \sum_{k=0}^{K-1} \mathbb{E}_k |J_c^{\theta_k} - \eta_{c,k,\text{MLMC}}|^2 \\
&\stackrel{(a)}{\leq} \beta + \frac{2}{\delta K} \sum_{k=0}^{K-1} |J_c^{\theta_k} - \mathbb{E}_k[\eta_{c,k,\text{MLMC}}]| \\
&\quad + \frac{\beta}{2K} \sum_{k=0}^{K-1} \mathbb{E}_k |J_c^{\theta_k} - \eta_{c,k,\text{MLMC}}|^2 \\
&\stackrel{(b)}{\leq} \beta + \frac{2}{\delta K} \sum_{k=0}^{K-1} \|\xi_{c,k}^* - \mathbb{E}_k[\xi_{c,k,\text{MLMC}}]\| \\
&\quad + \frac{\beta}{2K} \sum_{k=0}^{K-1} \mathbb{E}_k \|\xi_{c,k}^* - \xi_{c,k,\text{MLMC}}\|^2, \quad (26)
\end{aligned}$$

where (a) follows since $|J_c^{\theta_k}| \leq 1$, $0 \leq \lambda_k \leq 2/\delta$, and (b) follows from the definitions of $\xi_{c,k}^*$ and $\xi_{c,k,\text{MLMC}}$.

Using the definition of $\mathcal{L}(\theta, \lambda)$ and $\mathcal{L}(\pi^*, \lambda)$, we have

$$\begin{aligned}
\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}[J_r^{\pi^*} - J_r^{\theta_k}] &= -\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}[\lambda_k (J_c^{\pi^*} - J_c^{\theta_k})] \\
&\quad + \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}[\mathcal{L}(\pi^*, \lambda_k) - \mathcal{L}(\theta_k, \lambda_k)].
\end{aligned}$$

We combine the above relation with (26) to complete the proof of Lemma 1. $\square$

B Bounding Constraint Violation Rate: Proof of Lemma 2

We have from the dual update in (15), that

$$\begin{aligned}
&\mathbb{E}_k \left(\lambda_{k+1} - \frac{2}{\delta} \right)^2 - \left(\lambda_k - \frac{2}{\delta} \right)^2 \\
&\stackrel{(a)}{\leq} \mathbb{E}_k \left(\lambda_k - \beta \eta_{c,k,\text{MLMC}} - \frac{2}{\delta} \right)^2 - \left(\lambda_k - \frac{2}{\delta} \right)^2 \\
&= \beta^2 \mathbb{E}_k \eta_{c,k,\text{MLMC}}^2 - 2\beta \mathbb{E}_k \left[\eta_{c,k,\text{MLMC}} \left(\lambda_k - \frac{2}{\delta} \right) \right] \\
&\leq 2\beta^2 \mathbb{E}_k |J_c^{\theta_k} - \eta_{c,k,\text{MLMC}}|^2 \\
&\quad + 2\beta \left(\lambda_k - \frac{2}{\delta} \right) (J_c^{\theta_k} - \mathbb{E}_k[\eta_{c,k,\text{MLMC}}]) \\
&\quad - 2\beta J_c^{\theta_k} \left(\lambda_k - \frac{2}{\delta} \right) + 2\beta^2 (J_c^{\theta_k})^2 \\
&\stackrel{(b)}{=} \mathcal{O} \left(\beta^2 \mathbb{E}_k \|\xi_{c,k}^* - \xi_{c,k,\text{MLMC}}\|^2 \right. \\
&\quad \left. + \beta \|\xi_{c,k}^* - \mathbb{E}_k[\xi_{c,k,\text{MLMC}}]\| - \beta J_c^{\theta_k} \left(\lambda_k - \frac{2}{\delta} \right) + \beta^2 \right),
\end{aligned}$$

where (a) follows from non-expansiveness of the projection $\Pi_{[0,2/\delta]}$, and (b) follows by using the definitions of $\xi_{c,k}^*$ and $\xi_{c,k,\text{MLMC}}$, and using $\lambda_k \leq 2/\delta$ and $J_c^{\theta_k} \leq 1$.

Summing both sides over $0 \leq k < K$, multiplying with β/K , and taking expectation gives us

$$\begin{aligned}
&\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \left[J_c^{\theta_k} \left(\lambda_k - \frac{2}{\delta} \right) \right] \\
&= \mathcal{O} \left(\beta + \frac{1}{\beta K} + \frac{\beta}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\xi_{c,k}^* - \xi_{c,k,\text{MLMC}}\|^2 \right. \\
&\quad \left. + \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\xi_{c,k}^* - \mathbb{E}_k[\xi_{c,k,\text{MLMC}}]\| \right). \quad (27)
\end{aligned}$$

Now, note that

$$\begin{aligned}
& \sum_{k=0}^{K-1} \left[J_r^{\pi^*} - J_r^{\theta_k} \right] - \frac{2}{\delta} \sum_{k=0}^{K-1} J_c^{\theta_k} \\
& \stackrel{(a)}{\leq} \sum_{k=0}^{K-1} \left[J_r^{\pi^*} - J_r^{\theta_k} \right] + \sum_{k=0}^{K-1} \left[J_c^{\pi^*} - \frac{2}{\delta} J_c^{\theta_k} \right] \\
& \stackrel{(b)}{=} \sum_{k=0}^{K-1} [\mathcal{L}(\pi^*, \lambda_k) - \mathcal{L}(\theta_k, \lambda_k)] + \sum_{k=0}^{K-1} \left(\lambda_k - \frac{2}{\delta} \right) J_c^{\theta_k}.
\end{aligned}$$

where (a) uses the fact that $J_c^{\pi^*} \geq 0$, while (b) uses the definition of the Lagrangian $\mathcal{L}$. Dividing both sides by K , taking expectation, and substituting the last term using (27) then gives us

$$\begin{aligned}
& J_r^{\pi^*} - \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [J_r^{\theta_k}] + \frac{2}{\delta K} \sum_{k=0}^{K-1} \mathbb{E} [-J_c^{\theta_k}] \\
& = \mathcal{O} \left(\beta + \frac{1}{\beta K} + \frac{\beta}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\xi_{c,k}^* - \xi_{c,k,\text{MLMC}}\|^2 \right. \\
& \quad + \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\xi_{c,k}^* - \mathbb{E}_k [\xi_{c,k,\text{MLMC}}]\| \\
& \quad \left. + \frac{1}{K} \sum_{k=0}^{K-1} [\mathcal{L}(\pi^*, \lambda_k) - \mathcal{L}(\theta_k, \lambda_k)] \right). \quad (28)
\end{aligned}$$

We define a policy $\bar{\pi}_K$ that picks a policy from $\{\pi_{\theta_k}\}_{k < K}$, uniformly at random. Then, (28) implies

$$\begin{aligned}
& J_r^{\pi^*} - \mathbb{E} [J_r^{\bar{\pi}_K}] + \frac{2}{\delta} \mathbb{E} [-J_c^{\bar{\pi}_K}] \\
& = \mathcal{O} \left(\beta + \frac{1}{\beta K} + \frac{\beta}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\xi_{c,k}^* - \xi_{c,k,\text{MLMC}}\|^2 \right. \\
& \quad + \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\xi_{c,k}^* - \mathbb{E}_k [\xi_{c,k,\text{MLMC}}]\| \\
& \quad \left. + \frac{1}{K} \sum_{k=0}^{K-1} [\mathcal{L}(\pi^*, \lambda_k) - \mathcal{L}(\theta_k, \lambda_k)] \right).
\end{aligned}$$

By (Xu et al. 2025, Lemma G.6), we conclude

$$\begin{aligned}
& \mathbb{E} [-J_c^{\bar{\pi}_K}] = \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [-J_c^{\theta_k}] \\
& = \mathcal{O} \left(\frac{\delta}{\beta K} + \delta \beta + \frac{\delta \beta}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\xi_{c,k}^* - \xi_{c,k,\text{MLMC}}\|^2 \right. \\
& \quad + \frac{\delta}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\xi_{c,k}^* - \mathbb{E}_k [\xi_{c,k,\text{MLMC}}]\| \\
& \quad \left. + \frac{\delta}{K} \sum_{k=0}^{K-1} [\mathcal{L}(\pi^*, \lambda_k) - \mathcal{L}(\theta_k, \lambda_k)] \right).
\end{aligned}$$

This completes the proof of Lemma 2. $\square$

C Lagrange Error Analysis: Proof of Lemma 3

Let expectation w.r.t stationary distributions d^{π^*} or ν^{π^*} be denoted by $\mathbb{E}_*[\cdot]$. Then, we have for $0 \leq k < K$,

$$\begin{aligned}
& \mathbb{E}_* [\text{KL}(\pi^*(\cdot|s) || \pi_{\theta_k}(\cdot|s)) - \text{KL}(\pi^*(\cdot|s) || \pi_{\theta_{k+1}}(\cdot|s))] \\
& = \mathbb{E}_* \left[\ln \frac{\pi_{\theta_{k+1}}(a|s)}{\pi_{\theta_k}(a|s)} \right] \\
& \stackrel{(a)}{\geq} \mathbb{E}_* \langle \nabla_{\theta} \ln \pi_{\theta_k}(a|s), \theta_{k+1} - \theta_k \rangle - \frac{G_2}{2} \|\theta_{k+1} - \theta_k\|^2 \\
& \stackrel{(b)}{=} \alpha \mathbb{E}_* \langle \nabla_{\theta} \ln \pi_{\theta_k}(a|s), w_k \rangle - \frac{G_2 \alpha^2}{2} \|w_k\|^2 \\
& \geq \alpha \mathbb{E}_* \langle \nabla_{\theta} \ln \pi_{\theta_k}(a|s), w_k \rangle - \frac{G_2 \alpha^2}{2} \|w_k\|^2 \\
& \stackrel{(c)}{=} \alpha [\mathcal{L}(\pi^*, \lambda_k) - \mathcal{L}(\theta_k, \lambda_k)] - \frac{G_2 \alpha^2}{2} \|w_k\|^2 \\
& + \alpha \mathbb{E}_* \langle \nabla_{\theta} \ln \pi_{\theta_k}(a|s), w_k \rangle \\
& \quad - \alpha [J_r^{\pi^*} - J_r^{\theta_k}] - \alpha \lambda_k [J_c^{\pi^*} - J_c^{\theta_k}] \\
& \stackrel{(d)}{=} \alpha [\mathcal{L}(\pi^*, \lambda_k) - \mathcal{L}(\theta_k, \lambda_k)] - \frac{G_2 \alpha^2}{2} \|w_k\|^2 \\
& + \alpha \mathbb{E}_* \langle \nabla_{\theta} \ln \pi_{\theta_k}(a|s), w_k - w_k^* \rangle \\
& + \alpha \mathbb{E}_* [\langle \nabla_{\theta} \ln \pi_{\theta_k}(a|s), w_k^* \rangle - A_r^{\theta_k}(s, a) - \lambda_k A_c^{\theta_k}(s, a)] \\
& \stackrel{(e)}{\geq} \alpha [\mathcal{L}(\pi^*, \lambda_k) - \mathcal{L}(\theta_k, \lambda_k)] - \frac{G_2 \alpha^2}{2} \|w_k\|^2 - \alpha \sqrt{\varepsilon_{\text{bias}}}, \\
& + \alpha \mathbb{E}_* \langle \nabla_{\theta} \ln \pi_{\theta_k}(a|s), w_k - w_k^* \rangle,
\end{aligned}$$

where (a) follows from Assumption 4, (b) follows from (15), (c) follows from (3), (d) follows from Lemma 5, and (e) follows from combining Assumption 6 with the fact that $x \mapsto \sqrt{x}$ is concave and Assumption 6.

Now, rearranging terms and taking the conditional expectation w.r.t $\mathcal{F}_k$ gives

$$\begin{aligned}
& \mathbb{E}_k [\mathcal{L}(\pi^*, \lambda_k) - \mathcal{L}(\theta_k, \lambda_k)] - \sqrt{\varepsilon_{\text{bias}}} - \frac{1}{\alpha} \\
& \times \mathbb{E}_* [\text{KL}(\pi^*(\cdot|s) || \pi_{\theta_k}(\cdot|s)) - \text{KL}(\pi^*(\cdot|s) || \pi_{\theta_{k+1}}(\cdot|s))] \\
& \stackrel{(a)}{\leq} -\langle \mathbb{E}_* [\nabla_{\theta} \ln \pi_{\theta_k}(a|s)], \mathbb{E}_k [w_k] - w_k^* \rangle + \frac{G_2 \alpha}{2} \|w_k\|^2 \\
& \stackrel{(b)}{\leq} G_1 \|\mathbb{E}_k [w_k] - w_k^*\| + G_2 \alpha (\mathbb{E} \|w_k - w_k^*\|^2 + \mathbb{E} \|w_k^*\|^2) \\
& \stackrel{(c)}{\leq} G_1 \|\mathbb{E}_k [w_k] - w_k^*\| + G_2 \alpha \mathbb{E} \|w_k - w_k^*\|^2 + \mathcal{O}(\tau_{\text{mix}}^2 \alpha),
\end{aligned}$$

where (a) follows since θ_k is $\mathcal{F}_k$ -measurable and w_k is independent of ν^{π^*} , (b) uses the inequality $(a+b)^2 \leq 2a^2 + 2b^2$, while (c) follows as we use Assumptions 2, 4, and 5 to get

$$\begin{aligned}
\|w_k^*\| & = \|F(\theta_k)^{\dagger} \nabla_{\theta} \mathcal{L}(\theta_k, \lambda_k)\| \\
& \leq \frac{1}{\mu_F} \left(\|\nabla_{\theta} J_r^{\theta_k}\| + \frac{2}{\delta} \|\nabla_{\theta} J_c^{\theta_k}\| \right),
\end{aligned}$$

and use Assumption 1 to conclude

$$\|\nabla_{\theta} J_u^{\theta_k}\| \leq G_1 \mathbb{E}_{\pi_{\theta_k}} \|A_u^{\theta_k}(s, a)\| = \mathcal{O}(\tau_{\text{mix}}).$$

Finally, the result follows by summing both sides over $0 \leq k < K$, dividing by K , using the fact that the KL-divergence is non-negative, and taking the expectation. $\square$

D NPG Estimator Analysis: Proof of

Theorem 2

Throughout the subsequent sections, we let $\mathbb{E}_{k,0}[\cdot]$ denote the conditional expectation $\mathbb{E}_k[\cdot]$. Further, for $h > 0$, we let $\mathbb{E}_{k,h}[\cdot]$ denote the conditional expectation w.r.t $\mathcal{F}_{k,h}$, i.e., all samples obtained upto the k -th outer-loop iteration and the subsequent $(h-1)$ inner-loop iterations.

Recall from (24), that the NPG update is given by the following linear stochastic recursion:

$$w_{h+1}^k = w_h^k + \gamma_w \left[\left(\hat{g}_{r,\text{MLMC}}^{k,h} + \lambda_k \hat{g}_{c,\text{MLMC}}^{k,h} \right) - \hat{F}_{\text{MLMC}}^{k,h} w_h^k \right], \quad (29)$$

where $(\hat{g}_{r,\text{MLMC}}^{k,h} + \lambda_k \hat{g}_{c,\text{MLMC}}^{k,h})$ and $\hat{F}_{\text{MLMC}}^{k,h}$ are MLMC-based estimators for the Lagrangian gradient $\nabla_{\theta} \mathcal{L}(\theta_k, \lambda_k)$ and the Fisher matrix $F(\theta_k)$, respectively.

Note that, under Assumption 5, w_k^* is the unique vector satisfying $F(\theta_k)w_k^* = \nabla_{\theta} \mathcal{L}(\theta_k, \lambda_k)$. We claim that the NPG update satisfies the conditions needed in Lemma 7, which bounds the convergence rate achieved by a general linear stochastic recursion.

Note that, by Assumptions 4 and 5, for all $\theta \in \Theta$ and $\lambda \in [0, 2/\delta]$,

$$\mu_F \leq \|F(\theta)\| \leq G_1^2$$

$$\text{and } \|\nabla_{\theta} \mathcal{L}(\theta, \lambda)\| \leq \left(1 + \frac{2}{\delta}\right) 4G_1 \tau_{\text{mix}}.$$

Next, we bound the bias and MSE of MLMC estimators $\hat{g}_{u,\text{MLMC}}^{k,h}$ and $\hat{F}_{\text{MLMC}}^{k,h}$. Equation (22) shows the naive estimator for $F(\theta_k)$ as $\frac{1}{T} \sum_{t=0}^{T-1} \hat{F}^k(s_t^{kh}, a_t^{kh}) = \nabla_{\theta} \ln \pi_{\theta_k}(a|s) (\nabla_{\theta} \ln \pi_{\theta_k}(a|s))^{\top}$. From Assumption 4, we have

$$\left\| \mathbb{E}_{\theta_k} \left[\hat{F}^{kh}(s, a) \right] - F(\theta_k) \right\| = 0$$

$$\text{and } \left\| \hat{F}^{kh}(s_t^{kh}, a_t^{kh}) - F(\theta_k) \right\|^2 \leq 4G_1^4.$$

Thus, by Lemma 8, we we have the bias and MSE of $\hat{F}_{\text{MLMC}}^{k,h}$ as

$$\delta_F^2 := \left\| \mathbb{E}_{kh} \left[\hat{F}_{\text{MLMC}}^{kh} \right] - F(\theta_k) \right\|^2 = \mathcal{O}(\tau_{\text{mix}}/T_{\text{max}}),$$

$$\sigma_F^2 := \mathbb{E}_{kh} \left\| \hat{F}_{\text{MLMC}}^{kh} - F(\theta_k) \right\|^2 = \mathcal{O}(\tau_{\text{mix}} \ln T_{\text{max}}). \quad (30)$$

Likewise, we bound the MSE and bias in $\hat{g}_{u,\text{MLMC}}^{k,h}$, recall from (22) that the naive estimator for $\nabla_{\theta} J_{\theta_k}^{\theta_k}$ is given by $\frac{1}{T} \sum_{t=0}^{T-1} \hat{g}_u^{k,h}(s_t^{kh}, a_t^{kh}, \bar{a}_t^{kh}, s_{t+1}^{kh}, a_{t+1}^{kh})$, where

$$\hat{g}_u^{k,h}(s, a, \bar{a}, s', a') = \left(u(s, a) - \eta_{u,k} + Q_{u,k,\text{MLMC}}(s', a') - Q_{u,k,\text{MLMC}}(s, \bar{a}) \right) \nabla_{\theta} \ln \pi_{\theta}(a|s).$$

Here, $Q_{u,k,\text{MLMC}}$ denotes the critic estimator for π_{θ_k} defined in (17).

Recall from (14) that Q_{lin} be the first-order Taylor approximation to $Q(\cdot, \cdot, \zeta)$. Then, we have

$$\begin{aligned} & \left\| \hat{g}_u^{k,h}(s, a, \bar{a}, s', a') \right\| \\ & \leq G_1 \left(|u(s, a)| + |\eta_{u,k,\text{MLMC}}| + 2\|Q_{u,k,\text{MLMC}}\| \right) \\ & = \mathcal{O} \left(|\eta_{u,k,\text{MLMC}}| + \|Q_{\text{lin}}(\cdot, \cdot, \zeta_{u,k,\text{MLMC}})\| \right. \\ & \quad \left. + \|Q_{u,k,\text{MLMC}} - Q_{\text{lin}}(\cdot, \cdot, \zeta_{u,k,\text{MLMC}})\| \right) \\ & = \mathcal{O} \left(|\eta_{u,k,\text{MLMC}}| + \|\zeta_{u,k,\text{MLMC}}\| \right. \\ & \quad \left. + \|Q_{u,k,\text{MLMC}} - Q_{\text{lin}}(\cdot, \cdot, \zeta_{u,k,\text{MLMC}})\| \right). \end{aligned}$$

Thus, the single-sample MSE for the naive PG estimator is

$$\begin{aligned} & \mathbb{E}_k \left\| \hat{g}_u^{k,h}(s_t^{kh}, a_t^{kh}, \bar{a}_t^{kh}, s_{t+1}^{kh}, a_{t+1}^{kh}) - \nabla_{\theta} J_{\theta_k}^{\theta_k} \right\|^2 \\ & = \tilde{\mathcal{O}} \left(\sigma_Q^2 + \|\xi_{u,k,\text{MLMC}} - \xi_{u,k}^*\|^2 \right), \end{aligned}$$

where we define

$$\sigma_Q^2 := \mathbb{E}_k \|Q_{u,k,\text{MLMC}} - Q_{\text{lin}}(\cdot, \cdot, \zeta_{u,k,\text{MLMC}})\|^2.$$

Now, we bound the single-sample bias for the naive PG estimator. Note that,

$$\begin{aligned} \mathbb{E}_{\theta_k} \left[\hat{g}_u^{k,h}(s, a, \bar{a}, s', a') \right] &= \mathbb{E}_{\theta_k} \left[\nabla_{\theta} \ln \pi_{\theta_k}(s, a) \right. \\ & \quad \times \left[(u(s, a) - J_u^{\theta_k}) + Q_u^{\theta_k}(s', a') - Q_u^{\theta_k}(s, \bar{a}) \right. \\ & \quad + (J_u^{\theta_k} - \eta_{u,k,\text{MLMC}}) + Q_u^{\theta_k}(s, \bar{a}) - Q_{\text{lin}}(s, \bar{a}, \zeta_{u,k}^*) \\ & \quad - (Q_u^{\theta_k}(s', a') - Q_{\text{lin}}(s', a', \zeta_{u,k}^*)) \\ & \quad + (Q_{\text{lin}}(s', a', \zeta_{u,k,\text{MLMC}}) - Q_{\text{lin}}(s', a', \zeta_{u,k}^*)) \\ & \quad - (Q_{\text{lin}}(s, \bar{a}, \zeta_{u,k,\text{MLMC}}) - Q_{\text{lin}}(s, \bar{a}, \zeta_{u,k}^*)) \\ & \quad + Q_{u,k,\text{MLMC}}(s, \bar{a}) - Q_{\text{lin}}(s, \bar{a}, \zeta_{u,k,\text{MLMC}}) \\ & \quad \left. \left. - (Q_{u,k,\text{MLMC}}(s', a') - Q_{\text{lin}}(s', a', \zeta_{u,k,\text{MLMC}})) \right] \right]. \quad (31) \end{aligned}$$

Applying the NTK bounds from Lemma 6 on (31) and using (8), we have the single-sample PG bias as

$$\begin{aligned} & \left\| \mathbb{E}_{\theta_k} \left[\hat{g}_u^{k,h}(s, a, \bar{a}, s', a') \right] - \nabla_{\theta} J_{\theta_k}^{\theta_k} \right\|^2 \\ & = \tilde{\mathcal{O}} \left(\varepsilon_{\text{app}} + \frac{1}{m} + \|\xi_{u,k,\text{MLMC}} - \xi_{u,k}^*\|^2 \right. \\ & \quad \left. + \|Q_{u,k,\text{MLMC}} - Q_{\text{lin}}(\cdot, \cdot, \zeta_{u,k,\text{MLMC}})\|^2 \right) \end{aligned}$$

and

$$\begin{aligned} & \left\| \mathbb{E}_k \mathbb{E}_{\theta_k} \left[\hat{g}_u^{k,h}(s, a, \bar{a}, s', a') \right] - \nabla_{\theta} J^{\theta_k} \right\|^2 \\ &= \tilde{\mathcal{O}} \left(\varepsilon_{\text{app}} + \frac{1}{m} + \left\| \mathbb{E}_k [\xi_{u,k,\text{MLMC}}] - \xi_{u,k}^* \right\|^2 \right. \\ & \quad \left. + \left\| \mathbb{E}_k [Q_{u,k,\text{MLMC}} - Q_{\text{lin}}(\cdot, \cdot, \zeta_{u,k,\text{MLMC}})] \right\|^2 \right). \end{aligned}$$

Consequently, applying Lemma 8, we get the following MSE and bias for $\hat{g}_{u,\text{MLMC}}^{k,h}$:

$$\begin{aligned} \sigma_g^2 &:= \sum_u \mathbb{E}_k \left\| \hat{g}_{u,\text{MLMC}}^{k,h} - \nabla_{\theta} J_u^{\theta_k} \right\|^2 \\ &= \tilde{\mathcal{O}} \left(\varepsilon_{\text{app}} + \frac{1}{m} + \sigma_Q^2 + \sum_u \left\| \xi_{u,k,\text{MLMC}} - \xi_{u,k}^* \right\|^2 \right), \end{aligned} \quad (32)$$

and

$$\begin{aligned} \delta_g^2 &:= \sum_u \left\| \mathbb{E}_{k,h} [\hat{g}_{u,\text{MLMC}}^{k,h}] - \nabla_{\theta} J_u^{\theta_k} \right\|^2 \\ &= \tilde{\mathcal{O}} \left(\varepsilon_{\text{app}} + \frac{1}{m} + \sigma_Q^2 + \sum_u \left\| \xi_{u,k,\text{MLMC}} - \xi_{u,k}^* \right\|^2 \right), \\ \bar{\delta}_g^2 &:= \sum_u \left\| \mathbb{E}_k [\hat{g}_{u,\text{MLMC}}^{k,h}] - \nabla_{\theta} J_u^{\theta_k} \right\|^2 \\ &= \tilde{\mathcal{O}} \left(\varepsilon_{\text{app}} + \frac{1}{m} + \delta_Q^2 + \sum_u \left\| \mathbb{E}_k [\xi_{u,k,\text{MLMC}}] - \xi_{u,k}^* \right\|^2 \right), \end{aligned} \quad (33)$$

where the sum is over $u \in \{r, c\}$ and

$$\delta_Q^2 := \left\| \mathbb{E}_k [Q_{u,k,\text{MLMC}} - Q_{\text{lin}}(\cdot, \cdot, \zeta_{u,k,\text{MLMC}})] \right\|^2.$$

Using these bounds, we are ready to apply Lemma 7 to the NPG update (29). Choosing the NPG stepsize $\gamma_w := \frac{2 \ln T}{\mu_F H_w}$, we have for T sufficiently large:

$$\begin{aligned} \mathbb{E}_k \|w_k - w_k^*\|^2 &= \tilde{\mathcal{O}} \left(\frac{1}{T^2} + \frac{1}{H_w} (\sigma_F^2 + \sigma_g^2) + \delta_F^2 + \delta_g^2 \right), \\ \left\| \mathbb{E}_k [w_k] - w_k^* \right\|^2 &= \tilde{\mathcal{O}} \left(\frac{1}{T^2} + \delta_F^2 + \bar{\delta}_g^2 \right. \\ & \quad \left. + \delta_P^2 \left(\frac{1}{H_w} [\sigma_F^2 + \sigma_g^2] + \delta_F^2 + \delta_g^2 \right) \right). \end{aligned}$$

Finally, substituting the bias and MSE terms in the above relations with (30), (32), and (33) completes the proof of Theorem 2. $\square$

E Critic Estimator Analysis: Proofs of Lemma 4 and Theorem 3

Proof of Lemma 4. We begin by writing the vanilla-critic update as a perturbed linear recursion. Let z, z' denote the state action pairs $(s, a), (s', a')$, respectively. Further, let $\psi(z)$ denote $\nabla_{\zeta} Q(z, \zeta_0)$. Then, replacing the neural critic

$Q(\cdot, \cdot, \zeta)$ with the linear critic $Q_{\text{lin}}(\cdot, \cdot, \zeta)$ in (19), we decompose the naive single-sample critic-gradient estimator as

$$\begin{aligned} \hat{v}^{k,h}(z, z') &:= \left[\hat{A}^{k,h}(z, z') \xi_{u,h}^k - \hat{B}_u^{k,h}(z, z') \right] \\ &\quad - \tilde{V}_u^{k,h}(z, z'), \end{aligned} \quad (34)$$

where

$$\begin{aligned} \hat{A}^{k,h}(z, z') &:= \begin{bmatrix} c_{\gamma} & 0 \\ \psi_z & \psi(z) [\psi(z) - \psi(z')]^{\top} \end{bmatrix}, \\ \hat{B}_u^{k,h}(z, z') &:= \begin{bmatrix} c_{\gamma} u(z) \\ (u(z) - Q(z, \zeta_0) + Q(z', \zeta_0)) \psi(z) \end{bmatrix}, \end{aligned}$$

and $\tilde{V}_u^{k,h}(z, z')$ collects the remainder terms. This decomposition then translates to the following decomposition of the MLMC critic-gradient estimator in (20):

$$\hat{v}_{\text{MLMC}}^{k,h} := \left[\hat{A}_{\text{MLMC}}^{k,h} \xi_{u,h}^k - \hat{B}_{u,\text{MLMC}}^{k,h} \right] - \tilde{V}_{u,\text{MLMC}}^{k,h}, \quad (35)$$

where $\hat{A}_{\text{MLMC}}^{k,h}$, $\hat{B}_{u,\text{MLMC}}^{k,h}$, and $\tilde{V}_{u,\text{MLMC}}^{k,h}$ are obtained by replacing $\hat{v}_u^{k,h}$ in (20) with $\hat{A}_u^{k,h}$, $\hat{B}_u^{k,h}$, and $\tilde{V}_u^{k,h}$, respectively.

With the decomposition (35), the vanilla-critic update (21) can be written as the following perturbed linear recursion:

$$\begin{aligned} \xi_{u,h+1}^k &= \Pi_B \left(\xi_{u,h}^k + \gamma_{\xi,H} [\hat{B}_{u,\text{MLMC}}^{k,h} - \hat{A}_{\text{MLMC}}^{k,h} \xi_{u,h}^k] \right. \\ & \quad \left. + \gamma_{\xi,H} \tilde{V}_{u,\text{MLMC}}^{k,h} \right). \end{aligned} \quad (36)$$

Recall that $\xi_{u,k}^*$ is the unique solution to $A(\theta_k) \xi = B_u(\theta_k)$, where

$$\begin{aligned} A(\theta_k) &:= \mathbb{E}_{\theta_k} [\hat{A}^{k,h}(z, z')], \\ \text{and } B_u(\theta_k) &:= \mathbb{E}_{\theta_k} [\hat{B}_u^{k,h}(z, z')]. \end{aligned} \quad (37)$$

Let $\hat{A}_{\text{MLMC}}^{k,h}$, $\hat{B}_{u,\text{MLMC}}^{k,h}$, and $\tilde{V}_{u,\text{MLMC}}^{k,h}$ satisfy the following MSE bounds

$$\begin{aligned} \mathbb{E}_{k,h} \|\hat{A}_{\text{MLMC}}^{k,h} - A(\theta_k)\|^2 &\leq \sigma_A^2, \\ \mathbb{E}_{k,h} \|\hat{B}_{u,\text{MLMC}}^{k,h} - B_u(\theta_k)\|^2 &\leq \sigma_B^2, \\ \text{and } \|\tilde{V}_{u,\text{MLMC}}^{k,h}\|^2 &\leq \sigma_V^2, \end{aligned} \quad (38)$$

and the following bias bounds

$$\begin{aligned} \left\| \mathbb{E}_{k,h} [\hat{A}_{\text{MLMC}}^{k,h}] - A(\theta_k) \right\|^2 &\leq \delta_A^2, \\ \text{and } \left\| \mathbb{E}_{k,h} [\hat{B}_{u,\text{MLMC}}^{k,h}] - B_u(\theta_k) \right\|^2 &\leq \delta_B^2. \end{aligned} \quad (39)$$

If we define the auxiliary linear recursion

$$\begin{aligned} \chi_{u,h+1}^k &= \Pi_B \left(\chi_{u,h}^k + \gamma_{\xi,H} [\hat{B}_{u,\text{MLMC}}^{k,h} - \hat{A}_{\text{MLMC}}^{k,h} \chi_{u,h}^k] \right), \end{aligned} \quad (40)$$

where $\chi_{u,0}^k = \xi_{u,0}^k$, then subtracting (36) from (40) gives us

$$\begin{aligned}
& \|\xi_{u,h+1}^k - \chi_{u,h+1}^k\|^2 \\
& \stackrel{(a)}{\leq} \left\| (I - \gamma_{\xi,H} \hat{A}_{\text{MLMC}}^{k,h}) (\xi_{u,h}^k - \chi_{u,h}^k) - \gamma_{\xi,H} \tilde{V}_{u,\text{MLMC}}^{k,h} \right\|^2 \\
& = \left\| (I - \gamma_{\xi,H} A(\theta_k)) (\xi_{u,h}^k - \chi_{u,h}^k) \right. \\
& \quad \left. + \gamma_{\xi,H} (A(\theta_k) - \hat{A}_{\text{MLMC}}^{k,h}) - \gamma_{\xi,H} \tilde{V}_{u,\text{MLMC}}^{k,h} \right\|^2 \\
& \stackrel{(a)}{=} \left\| (I - \gamma_{\xi,H} A(\theta_k)) (\xi_{u,h}^k - \chi_{u,h}^k) \right\|^2 \\
& \quad + \gamma_{\xi,H}^2 \left\| A(\theta_k) - \hat{A}_{\text{MLMC}}^{k,h} \right\|^2 + \gamma_{\xi,H}^2 \left\| \tilde{V}_{u,\text{MLMC}}^{k,h} \right\|^2 \\
& \quad - 2\gamma_{\xi,H}^2 \left\langle A(\theta_k) - \hat{A}_{\text{MLMC}}^{k,h}, \tilde{V}_{u,\text{MLMC}}^{k,h} \right\rangle \\
& \quad + 2\gamma_{\xi,H} \left\langle A(\theta_k) - \hat{A}_{\text{MLMC}}^{k,h}, \right. \\
& \quad \quad \left. \times (I - \gamma_{\xi,H} A(\theta_k)) (\xi_{u,h}^k - \chi_{u,h}^k) \right\rangle, \quad (41)
\end{aligned}$$

where (a) follows from the non-expansiveness of Π_B . Taking conditional expectation on both sides, we have

$$\begin{aligned}
& \mathbb{E}_{k,h} \|\xi_{u,h+1}^k - \chi_{u,h+1}^k\|^2 \\
& \stackrel{(a)}{=} \left\| (I - \gamma_{\xi,H} A(\theta_k)) (\xi_{u,h}^k - \chi_{u,h}^k) \right\|^2 \\
& \quad + \gamma_{\xi,H}^2 \mathbb{E}_{k,h} \left\| A(\theta_k) - \hat{A}_{\text{MLMC}}^{k,h} \right\|^2 + \gamma_{\xi,H}^2 \mathbb{E}_{k,h} \left\| \tilde{V}_{u,\text{MLMC}}^{k,h} \right\|^2 \\
& \quad - 2\gamma_{\xi,H}^2 \mathbb{E}_{k,h} \left\langle A(\theta_k) - \hat{A}_{\text{MLMC}}^{k,h}, \tilde{V}_{u,\text{MLMC}}^{k,h} \right\rangle \\
& \quad + 2\gamma_{\xi,H} \left\langle A(\theta_k) - \mathbb{E}_{k,h} [\hat{A}_{\text{MLMC}}^{k,h}], \right. \\
& \quad \quad \left. \times (I - \gamma_{\xi,H} A(\theta_k)) (\xi_{u,h}^k - \chi_{u,h}^k) \right\rangle \\
& \stackrel{(b)}{\leq} (1 - \mu_\phi \gamma_{\xi,H})^2 \|\xi_{u,h}^k - \chi_{u,h}^k\|^2 \\
& \quad + 2\gamma_{\xi,H}^2 \mathbb{E}_{k,h} \left\| A(\theta_k) - \hat{A}_{\text{MLMC}}^{k,h} \right\|^2 \\
& \quad + 2\gamma_{\xi,H}^2 \mathbb{E}_{k,h} \left\| \tilde{V}_{u,\text{MLMC}}^{k,h} \right\|^2 \\
& \quad + 2\gamma_{\xi,H} \left\langle A(\theta_k) - \mathbb{E}_{k,h} [\hat{A}_{\text{MLMC}}^{k,h}], \right. \\
& \quad \quad \left. \times (I - \gamma_{\xi,H} A(\theta_k)) (\xi_{u,h}^k - \chi_{u,h}^k) \right\rangle \\
& \stackrel{(c)}{\leq} (1 - \mu_\phi \gamma_{\xi,H}) \|\xi_{u,h}^k - \chi_{u,h}^k\|^2 + 2\gamma_{\xi,H}^2 (\sigma_A^2 + \sigma_V^2) \\
& \quad + 2\gamma_{\xi,H} (1 - \mu_\phi \gamma_{\xi,H}) \|\xi_{u,h}^k - \chi_{u,h}^k\| \delta_A \\
& \stackrel{(d)}{\leq} \exp(-\mu_\phi \gamma_{\xi,H}) \|\xi_{u,h}^k - \chi_{u,h}^k\|^2 \\
& \quad + \mathcal{O}(\gamma_{\xi,H}^2 (\sigma_A^2 + \sigma_V^2) + \gamma_{\xi,H} R \delta_A), \quad (42)
\end{aligned}$$

where (a) follows since $\xi_{u,h}^k, \chi_{u,h}^k$ are $\mathcal{F}_{k,h}$ -measurable, (b) follows since assuming $\ker(A(\theta_k)) = \{0\}$, (this holds if the vectors $\{\nabla_\zeta Q(s, a, \zeta_0)\}$ do not span the all ones vector in $\mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$) we can show that for c_γ sufficiently large, $\xi^\top A(\theta_k) \xi \geq \mu_\phi \|\xi\|^2$ (see (Satheesh et al. 2026, Lemma 17)), and using Cauchy-Schwarz inequality, and (c) follows by choosing $\gamma_{\xi,H} < \mu_\phi/2$ and using the bounds (38) and

(39). Finally, (d) follows since $\xi_{u,H}^k, \chi_{u,H}^k$ lie in the NTK ball and by using the inequality $1 - x < e^{-x}$.

Then, applying (42) recursively and following the arguments in the proof of (Xu et al. 2025, Theorem B.1), we get

$$\begin{aligned}
\mathbb{E}_k \|\xi_{u,H}^k - \chi_{u,H}^k\|^2 & = \mathcal{O} \left(\mathbb{E}_k \|\xi_{u,0}^k - \chi_{u,0}^k\|^2 e^{-\mu_\phi H \gamma_{\xi,H}} \right. \\
& \quad \left. + \gamma_{\xi,H} (\sigma_A^2 + \sigma_V^2) + R \delta_A^2 \right). \quad (43)
\end{aligned}$$

Taking $R = \Theta(\ln T)$ and $\gamma_{\xi,H} = \frac{2 \ln T}{\mu_\phi H}$, and using the fact that $\xi_{u,0}^k = \chi_{u,0}^k$,

$$\mathbb{E}_k \|\xi_{u,H}^k - \chi_{u,H}^k\|^2 = \tilde{\mathcal{O}} \left(\frac{1}{H} (\sigma_A^2 + \sigma_V^2) + \delta_A^2 \right), \quad (44)$$

Taking the NTK radius R sufficiently large so that $\xi_{u,k}^*$ lies in the NTK ball, we apply the same recipe used to obtain (44) to the auxiliary update (40). This gives us

$$\begin{aligned}
\mathbb{E}_k \|\chi_{u,H}^k - \xi_{u,k}^*\|^2 & = \tilde{\mathcal{O}} \left(\frac{1}{H} (\sigma_A^2 + \sigma_B^2) + \delta_A^2 + \delta_B^2 + \sigma_V^2 \right). \quad (45)
\end{aligned}$$

Hence, to conclude Lemma 4, it remains to bound the MSEs and biases of $\hat{A}_{\text{MLMC}}^{k,h}$, $\hat{B}_{u,\text{MLMC}}^{k,h}$, and $\tilde{V}_{u,\text{MLMC}}^{k,h}$. By Lemma 6 and (37), the corresponding single-sample biases satisfy

$$\begin{aligned}
\|\mathbb{E}_{\theta_k} [\tilde{V}_{u,\text{MLMC}}^{k,h}(z, z')] - \tilde{V}_{u,\text{MLMC}}^{k,h}(z, z')\| & = \tilde{\mathcal{O}}(R m^{-1/2}), \\
\|\mathbb{E}_{\theta_k} [\hat{A}^{k,h}(z, z')] - A(\theta_k)\| & = 0, \\
\|\mathbb{E}_{\theta_k} [\hat{B}_{u,\text{MLMC}}^{k,h}(z, z')] - B_u(\theta_k)\| & = 0, \quad (46)
\end{aligned}$$

while under NTK regime, the single-sample MSEs satisfy

$$\begin{aligned}
\|\tilde{V}_{u,\text{MLMC}}^{k,h}(z, z') - \mathbb{E}_{\theta_k} [\tilde{V}_{u,\text{MLMC}}^{k,h}(z, z')]\| & = \tilde{\mathcal{O}}(R m^{-1/2}), \\
\|\hat{A}^{k,h}(z, z') - \mathbb{E}_{\theta_k} [\hat{A}^{k,h}(z, z')]\| & = \mathcal{O}(c_\gamma^2), \\
\|\hat{B}_{u,\text{MLMC}}^{k,h}(z, z') - \mathbb{E}_{\theta_k} [\hat{B}_{u,\text{MLMC}}^{k,h}(z, z')]\| & = \tilde{\mathcal{O}}(c_\gamma^2). \quad (47)
\end{aligned}$$

Applying Lemma 8, we conclude $\sigma_V^2 = \tilde{\mathcal{O}}(\tau_{\text{mix}} m^{-1})$ and

$$\begin{aligned}
\sigma_A^2 & = \mathcal{O}(\tau_{\text{mix}} \ln T_{\text{max}}), \quad \delta_A^2 = \mathcal{O}(\tau_{\text{mix}} T_{\text{max}}^{-1}), \\
\sigma_B^2 & = \mathcal{O}(\tau_{\text{mix}} \ln T_{\text{max}}), \quad \delta_B^2 = \mathcal{O}(\tau_{\text{mix}} T_{\text{max}}^{-1}). \quad (48)
\end{aligned}$$

Finally, we substitute the above bounds into (44) and (45), and use the inequality

$$\begin{aligned}
\mathbb{E}_k \|\xi_{u,H}^k - \xi_{u,k}^*\|^2 & \leq 2\mathbb{E}_k \|\xi_{u,H}^k - \chi_{u,H}^k\|^2 + 2\mathbb{E}_k \|\chi_{u,H}^k - \xi_{u,k}^*\|^2. \quad (49)
\end{aligned}$$

This completes the proof of Lemma 4. $\square$

Next, we conclude this section with the proof of Theorem 3.

Proof of Theorem 3. Recall from (16) and (17), that

$$\begin{aligned}
Q_{u,k,\text{MLMC}}(\cdot, \cdot) & := Q(\cdot, \cdot, \zeta_{u,1}^k) \\
& + \mathbb{1}_{\{2^{H_k} \leq H_{\text{max}}\}} 2^{H_k} \left[Q(\cdot, \cdot, \zeta_{u,2^{H_k}}^k) - Q(\cdot, \cdot, \zeta_{u,2^{H_k-1}}^k) \right]
\end{aligned}$$

and

$$\xi_{u,k,\text{MLMC}} := \xi_{u,1}^k + \mathbb{1}_{\{2^{H_k} \leq H_{\max}\}} 2^{H_k} \left(\xi_{u,2^{H_k}}^k - \xi_{u,2^{H_k-1}}^k \right),$$

where $H_k \sim \text{Geom}(1/2)$. Then, we have

$$\begin{aligned} & \mathbb{E}_k [Q_{u,k,\text{MLMC}}(\cdot, \cdot)] - \mathbb{E}_k [Q(\cdot, \cdot, \zeta_{u,1}^k)] \\ & \stackrel{(a)}{=} \sum_{h=1}^{\lfloor \log_2 H_{\max} \rfloor} 2^h \cdot \mathbb{P}(H_k = h) \\ & \quad \times \left(\mathbb{E}_k [Q(\cdot, \cdot, \zeta_{u,2^h}^k)] - \mathbb{E}_k [Q(\cdot, \cdot, \zeta_{u,2^{h-1}}^k)] \right) \\ & \stackrel{(b)}{=} \sum_{h=1}^{\lfloor \log_2 H_{\max} \rfloor} 2^h \cdot 2^{-h} \\ & \quad \times \left(\mathbb{E}_k [Q(\cdot, \cdot, \zeta_{u,2^h}^k)] - \mathbb{E}_k [Q(\cdot, \cdot, \zeta_{u,2^{h-1}}^k)] \right) \\ & = \sum_{h=1}^{\lfloor \log_2 H_{\max} \rfloor} \left(\mathbb{E}_k [Q(\cdot, \cdot, \zeta_{u,2^h}^k)] - \mathbb{E}_k [Q(\cdot, \cdot, \zeta_{u,2^{h-1}}^k)] \right) \\ & = \mathbb{E}_k [Q(\cdot, \cdot, \zeta_{u,2^{\lfloor \log_2 H_{\max} \rfloor}}^k)] - \mathbb{E}_k [Q(\cdot, \cdot, \zeta_{u,1}^k)], \quad (50) \end{aligned}$$

where (a) uses (16) and (b) follows since $H_k \sim \text{Geom}(1/2)$. Hence,

$$\mathbb{E}_k [Q_{u,k,\text{MLMC}}(\cdot, \cdot)] = \mathbb{E}_k [Q(\cdot, \cdot, \zeta_{u,2^{\lfloor \log_2 H_{\max} \rfloor}}^k)]. \quad (51)$$

A similar telescoping argument gives us

$$\mathbb{E}_k [\xi_{u,k,\text{MLMC}}] = \mathbb{E}_k [\xi_{u,2^{\lfloor \log_2 H_{\max} \rfloor}}^k]. \quad (52)$$

Thus, we have the critic-function bias δ_Q^2

$$\begin{aligned} & = \left\| \mathbb{E}_k [Q_{u,k,\text{MLMC}}(\cdot, \cdot)] - Q_{\text{lin}}(\cdot, \cdot, \zeta_{u,k,\text{MLMC}}) \right\|^2 \\ & \stackrel{(a)}{=} \left\| \mathbb{E}_k [Q(\cdot, \cdot, \zeta_{u,2^{\lfloor \log_2 H_{\max} \rfloor}}^k)] - Q_{\text{lin}}(\cdot, \cdot, \zeta_{u,2^{\lfloor \log_2 H_{\max} \rfloor}}^k) \right\|^2 \\ & \stackrel{(b)}{=} \tilde{\mathcal{O}}(1/m), \end{aligned}$$

where (a) follows from (51) and (52), together with the fact that $Q_{\text{lin}}(\cdot, \cdot, \zeta)$ is linear in ζ . Whereas, (b) follows from the NTK bounds in Lemma 6 since $\zeta_{u,2^{\lfloor \log_2 H_{\max} \rfloor}}$ is contained within the NTK ball $\mathcal{B}$.

Further, the critic-parameter bias satisfies

$$\begin{aligned} & \left\| \mathbb{E}_k [\xi_{u,k,\text{MLMC}}] - \xi_{u,k}^* \right\|^2 \\ & \stackrel{(a)}{=} \left\| \mathbb{E}_k [\xi_{u,2^{\lfloor \log_2 H_{\max} \rfloor}}^k] - \xi_{u,k}^* \right\|^2 \\ & \stackrel{(b)}{\leq} \mathbb{E}_k \left\| \xi_{u,2^{\lfloor \log_2 H_{\max} \rfloor}}^k - \xi_{u,k}^* \right\|^2 \\ & \stackrel{(c)}{=} \tilde{\mathcal{O}} \left(\frac{1}{T^2} + \frac{\tau_{\text{mix}}}{T_{\text{max}}} + \frac{\tau_{\text{mix}}}{2^{\lfloor \log_2 H_{\max} \rfloor}} + \frac{\tau_{\text{mix}}}{\sqrt{m}} \right) \\ & = \tilde{\mathcal{O}} \left(\frac{\tau_{\text{mix}}}{T_{\text{max}}} + \frac{\tau_{\text{mix}}}{H_{\text{max}}} + \frac{\tau_{\text{mix}}}{m} \right), \end{aligned}$$

where (a) follows from (52), (b) follows since $\|\cdot\|$ is convex, and (c) follows from Lemma 4.

Now, we bound the critic MSE. Using a telescoping sum as earlier, we have the critic parameter MSE as

$$\begin{aligned} & \mathbb{E}_k \left\| \xi_{u,k,\text{MLMC}} - \xi_{u,k}^* \right\|^2 \\ & \stackrel{(a)}{\leq} 2 \mathbb{E}_k \left\| \xi_{u,1}^k - \xi_{u,k}^* \right\|^2 + \sum_{h=1}^{\lfloor \log_2 H_{\max} \rfloor} 2 \cdot 4^h \cdot \mathbb{P}(H_k = h) \\ & \quad \times \mathbb{E}_k \left\| \xi_{u,2^h}^k - \xi_{u,2^{h-1}}^k \right\|^2 \\ & \stackrel{(b)}{\leq} 2 \mathbb{E}_k \left\| \xi_{u,1}^k - \xi_{u,k}^* \right\|^2 + \sum_{h=1}^{\lfloor \log_2 H_{\max} \rfloor} 2^{2h+1} \cdot 2^{-h} \\ & \quad \times \left(2 \mathbb{E}_k \left\| \xi_{u,2^h}^k - \xi_{u,k}^* \right\|^2 + 2 \mathbb{E}_k \left\| \xi_{u,2^{h-1}}^k - \xi_{u,k}^* \right\|^2 \right) \\ & = 2 \mathbb{E}_k \left\| \xi_{u,1}^k - \xi_{u,k}^* \right\|^2 + \sum_{h=1}^{\lfloor \log_2 H_{\max} \rfloor} 2^{h+2} \\ & \quad \times \left(\mathbb{E}_k \left\| \xi_{u,2^h}^k - \xi_{u,k}^* \right\|^2 + \mathbb{E}_k \left\| \xi_{u,2^{h-1}}^k - \xi_{u,k}^* \right\|^2 \right) \\ & \stackrel{(c)}{=} \tilde{\mathcal{O}} \left(1 + \sum_{h=1}^{\lfloor \log_2 H_{\max} \rfloor} \frac{2^h}{m} \right) = \tilde{\mathcal{O}} \left(\frac{H_{\max}}{m} \right), \quad (53) \end{aligned}$$

where (a) uses (17), (b) follows since $H_k \sim \text{Geom}(1/2)$, and (c) follows from Lemma 4.

Similar arguments show the following:

$$\begin{aligned} & \mathbb{E}_k \left\| Q_{u,k,\text{MLMC}} - Q_{\text{lin}}(\cdot, \cdot, \zeta_{u,k,\text{MLMC}}) \right\|^2 \\ & \stackrel{(a)}{\leq} 2 \mathbb{E}_k \left\| Q(\cdot, \cdot, \zeta_{u,1}^k) - Q_{\text{lin}}(\cdot, \cdot, \zeta_{u,k,\text{MLMC}}) \right\|^2 \\ & \quad + \sum_{h=1}^{\lfloor \log_2 H_{\max} \rfloor} \frac{2 \cdot 4^h}{2^h} \mathbb{E}_k \left\| Q(\cdot, \cdot, \zeta_{u,2^h}^k) - Q(\cdot, \cdot, \zeta_{u,2^{h-1}}^k) \right\|^2 \\ & = 2 \mathbb{E}_k \left\| Q(\cdot, \cdot, \zeta_{u,1}^k) - Q_{\text{lin}}(\cdot, \cdot, \zeta_{u,k,\text{MLMC}}) \right\|^2 \\ & \quad + \sum_{h=1}^{\lfloor \log_2 H_{\max} \rfloor} 2^{h+1} \mathbb{E}_k \left\| Q(\cdot, \cdot, \zeta_{u,2^h}^k) - Q(\cdot, \cdot, \zeta_{u,2^{h-1}}^k) \right\|^2 \\ & \stackrel{(b)}{\leq} 2 \mathbb{E}_k \left\| Q_{\text{lin}}(\cdot, \cdot, \zeta_{u,1}^k) - Q_{\text{lin}}(\cdot, \cdot, \zeta_{u,k,\text{MLMC}}) \right\|^2 \\ & \quad + \sum_{h=1}^{\lfloor \log_2 H_{\max} \rfloor} 2^{h+1} \mathbb{E}_k \left\| Q_{\text{lin}}(\cdot, \cdot, \zeta_{u,2^h}^k) - Q_{\text{lin}}(\cdot, \cdot, \zeta_{u,2^{h-1}}^k) \right\|^2 \\ & \quad + \tilde{\mathcal{O}} \left(\frac{1}{m} \right) \left(1 + \sum_{h=1}^{\lfloor \log_2 H_{\max} \rfloor} 2^{h+1} \right) \\ & \stackrel{(c)}{=} \tilde{\mathcal{O}} \left(\mathbb{E}_k \left\| \zeta_{u,1}^k - \zeta_{u,k,\text{MLMC}} \right\|^2 + \frac{H_{\max}}{m} \right. \\ & \quad \left. + \sum_{h=1}^{\lfloor \log_2 H_{\max} \rfloor} 2^{h+1} \mathbb{E}_k \left\| \zeta_{u,2^h}^k - \zeta_{u,2^{h-1}}^k \right\|^2 \right) \\ & \stackrel{(d)}{=} \tilde{\mathcal{O}} \left(\frac{H_{\max}}{m} + \sum_{h=1}^{\lfloor \log_2 H_{\max} \rfloor} \tilde{\mathcal{O}} \left(\frac{2^h}{m} \right) \right), \quad (54) \end{aligned}$$

where (a) is obtained from (16) and using $H_k \sim \text{Geom}(1/2)$, to obtain (b), for each $\zeta \in \{\zeta_{u,1}^k, \zeta_{u,2^h}^k, \zeta_{u,2^{h-1}}^k\}$, we replace $Q(\cdot, \cdot, \zeta)$ with $Q_{\text{lin}}(\cdot, \cdot, \zeta)$ and bound $\|Q(\cdot, \cdot, \zeta) - Q_{\text{lin}}(\cdot, \cdot, \zeta)\|$ using the NTK bounds in Lemma 6, (c) follows since $Q_{\text{lin}}(\cdot, \cdot, \zeta)$ is Lipschitz in ζ , and (d) uses (53) and Lemma 4. Therefore, we conclude that

$$\sigma_Q^2 = \tilde{\mathcal{O}}\left(\frac{H_{\max}}{m}\right).$$

This completes the proof of Theorem 3. $\square$

F Proof of Theorem 1

We finally combine the bounds derived in Theorems 2 and 3, and Lemmas 3, 2, and 1 to prove Theorem 1.

We choose $m = H_{\max}$ and $H_{\max} = T_{\max} = T$, as prescribed in Theorem 1. Then, Theorem 3 gives the following critic bounds:

$$\sigma_Q^2 = \tilde{\mathcal{O}}(1) \quad \text{and} \quad \delta_Q^2 = \tilde{\mathcal{O}}(1/T), \quad (55)$$

and

$$\begin{aligned} \mathbb{E}_k \|\xi_{u,k,\text{MLMC}} - \xi_{u,k}^*\|^2 &= \tilde{\mathcal{O}}(1), \\ \|\mathbb{E}_k [\xi_{u,k,\text{MLMC}}] - \xi_{u,k}^*\|^2 &= \tilde{\mathcal{O}}(\tau_{\text{mix}}/T). \end{aligned} \quad (56)$$

Plugging the bounds from (55) and (56) into Theorem 2 and taking $H_w = \Theta(\ln T)$, we get

$$\begin{aligned} \mathbb{E}_k \|w_k - w_k^*\|^2 &= \tilde{\mathcal{O}}(\tau_{\text{mix}} + \varepsilon_{\text{app}}), \\ \|\mathbb{E}_k [w_k] - w_k^*\|^2 &= \tilde{\mathcal{O}}(\varepsilon_{\text{app}} + \frac{\tau_{\text{mix}}}{T}). \end{aligned} \quad (57)$$

Next, we substitute the above NPG bias and MSE bounds into Lemma 3 and choose $\alpha = 1/\sqrt{K}$ and $K = \Theta(T)$ to get the following Lagrange error:

$$\begin{aligned} &\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}[\mathcal{L}(\pi^*, \lambda_k) - \mathcal{L}(\theta_k, \lambda_k)] \\ &= \mathcal{O}\left(\sqrt{\varepsilon_{\text{bias}}} + \sqrt{\varepsilon_{\text{app}}} + \tau_{\text{mix}}^2 \alpha + \frac{1}{\alpha K} + \frac{\sqrt{\tau_{\text{mix}}}}{\sqrt{T}}\right) \\ &= \tilde{\mathcal{O}}\left(\sqrt{\varepsilon_{\text{bias}}} + \sqrt{\varepsilon_{\text{app}}} + \frac{\tau_{\text{mix}}^2}{\sqrt{T}}\right). \end{aligned} \quad (58)$$

Finally, we plug the bounds from (58) and (56) into Lemma 1 and choose $\beta = 1/\sqrt{T}$. This gives us the following averaged error:

$$J_r^{\pi^*} - \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}[J_r^{\theta_k}] = \tilde{\mathcal{O}}\left(\sqrt{\varepsilon_{\text{bias}}} + \sqrt{\varepsilon_{\text{app}}} + \frac{\tau_{\text{mix}}^2}{\sqrt{T}}\right).$$

Likewise, plugging (58) and (56) into Lemma 2 gives the constraint violation rate:

$$-\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}[J_c^{\theta_k}] = \tilde{\mathcal{O}}\left(\sqrt{\varepsilon_{\text{bias}}} + \sqrt{\varepsilon_{\text{app}}} + \frac{\tau_{\text{mix}}^2}{\sqrt{T}}\right).$$

This concludes the proof of Theorem 1. $\square$

G Technical lemmas

The following lemma is the standard performance difference lemma (Bai, Mondal, and Aggarwal 2024b, Lemma 4) for average-reward MDPs.

Lemma 5 (Performance Difference Lemma). *For any two policies π_θ and $\pi_{\theta'}$, and $u \in \{r, c\}$, we have*

$$J_u^\pi - J_u^{\pi'} \leq \mathbb{E}_{(s,a) \sim \nu^\pi} [A_u^{\theta'}(s, a)],$$

where $A_u^{\theta'}$ is the advantage function of policy $\pi_{\theta'}$ as defined in (9).

The following lemma, proved in (Satheesh et al. 2026, Lemma 15), collects several useful bounds for neural critics under the NTK regime.

Lemma 6 (NTK bounds). *Choose $R = \mathcal{O}(\ln T)$ and let $\xi_{u,h}^k \in \mathcal{B}$, for all $0 \leq h < h_k$. Then, the following holds with probability at least $(1 - \varepsilon - 2Le^{-Cm})$:*

- (a) $\|\nabla_\zeta Q(\cdot, \cdot, \zeta)\| \leq C_1$; $|Q(\cdot, \cdot, \zeta)| = \mathcal{O}(\sqrt{\ln(h_k/\varepsilon)})$;
- (b) $\|Q(\cdot, \cdot, \zeta) - Q_{\text{lin}}(\cdot, \cdot, \zeta)\| = \mathcal{O}\left(\frac{\sqrt{\ln(h_k/\varepsilon)}}{\sqrt{m}}\right)$;
- (c) $\|\nabla_\zeta Q(\cdot, \cdot, \zeta) - \nabla_\zeta Q(\cdot, \cdot, \zeta_0)\| = \mathcal{O}\left(\frac{\sqrt{\ln(h_k/\varepsilon)}}{\sqrt{m}}\right)$;
- (d) $\|\tilde{V}_u^{k,h}(\cdot, \cdot, \cdot, \cdot)\| = \mathcal{O}\left(\frac{\sqrt{\ln(h_k/\varepsilon)}}{\sqrt{m}}\right)$.

The following lemma, proved in (Ganesh, Mondal, and Aggarwal 2025b, Theorem 2), characterizes the bias and MSE of a stochastic linear recursion.

Lemma 7 (Stochastic Linear Recursion). *Consider the linear stochastic recursion*

$$x_{h+1} = x_h + \gamma[\hat{q}_h - \hat{P}_h x_h],$$

where $(\hat{q}_h, \hat{P}_h)_{h=0}^{H-1}$ are noisy estimates for $(q, P) \in \mathbb{R}^d \times \mathbb{R}^{d \times d}$. The target is an x^* such that $Px^* = q$. Suppose that $(\hat{q}_h, \hat{P}_h)$ satisfies:

$$\begin{aligned} \mathbb{E}_h \|\hat{P}_h - P\|^2 &\leq \sigma_P^2, \quad \mathbb{E}_h \|\hat{q}_h - q\|^2 \leq \sigma_q^2, \\ \|\mathbb{E}_h[\hat{P}_h] - P\|^2 &\leq \delta_P^2, \quad \|\mathbb{E}_h[\hat{q}_h] - q\|^2 \leq \delta_q^2. \end{aligned}$$

Further, let

$$\begin{aligned} \|\mathbb{E}[\hat{q}_h] - q\|^2 &\leq \bar{\delta}_q^2, \quad \lambda_P \leq \|P\| \leq \Lambda_P, \\ \|\hat{q}_h\| &\leq \Lambda_q, \quad \text{and} \quad \gamma \leq \frac{\lambda_P}{(24\sigma_P^2 + 8\Lambda_P^2)}. \end{aligned}$$

Then, after H iterations, we have

$$\begin{aligned} \mathbb{E}\|x_H - x^*\|^2 &\leq \|x_0 - x^*\|^2 e^{-\gamma\lambda_P H} + \mathcal{O}(\gamma R_0 + R_1) \\ \|\mathbb{E}[x_H] - x^*\|^2 &\leq \|x_0 - x^*\|^2 (e^{-\bar{\beta}\lambda_P H} + \delta_P^2) \\ &\quad + \mathcal{O}(\delta_P^2 (\gamma R_0 + R_1) + \bar{R}_1), \end{aligned}$$

where $R_0 = \mathcal{O}(\sigma_P^2 + \sigma_q^2)$,

$$R_1 = \mathcal{O}(\delta_P^2 + \delta_q^2), \quad \text{and} \quad \bar{R}_1 = \mathcal{O}(\delta_P^2 + \bar{\delta}_q^2).$$

The following lemma is adapted for biased estimates from (Dorfman and Levy 2022, Lemma 3.1), and summarizes the computational cost, bias, and MSE guarantees of MLMC estimation.

Lemma 8 (MLMC bounds). *Let (Z_t) be a time-homogeneous ergodic Markov chain with a unique invariant distribution d_Z , and a mixing-time τ_{mix} . Assume that $F(x, Z)$ is an estimate of $F(x)$. Let for all $t \geq 0$,*

$$\begin{aligned} \|\mathbb{E}_{d_Z}[F(x, Z)] - F(x)\|^2 &\leq \delta^2 \\ \text{and } \mathbb{E}\|F(x, Z_t) - \mathbb{E}_{d_Z}[F(x, Z)]\|^2 &\leq \sigma^2. \end{aligned}$$

Further, let $Q \sim \text{Geom}(1/2)$. Then, the MLMC estimator defined as

$$g_{\text{MLMC}} := g_1 + \mathbb{1}_{2^Q \leq T_{\max}} \cdot 2^Q (g_{2^Q} - g_{2^{Q-1}}),$$

where $g_j := j^{-1} \sum_{t=0}^{j-1} F(x, Z_t)$, satisfies the following bounds.

- (a) $\mathbb{E}[g_{\text{MLMC}}] = \mathbb{E}[g_{2^{\lfloor \log_2 T_{\max} \rfloor}}]$;
- (b) $\mathbb{E}\|F(x) - g_{\text{MLMC}}\|^2 = \mathcal{O}(\sigma^2 \tau_{\text{mix}} \log_2 T_{\max} + \delta^2)$;
- (c) $\|F(x) - \mathbb{E}[g_{\text{MLMC}}]\|^2 = \mathcal{O}(\sigma^2 \tau_{\text{mix}} T_{\max}^{-1} + \delta^2)$.